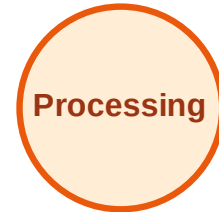


BFS Traversal

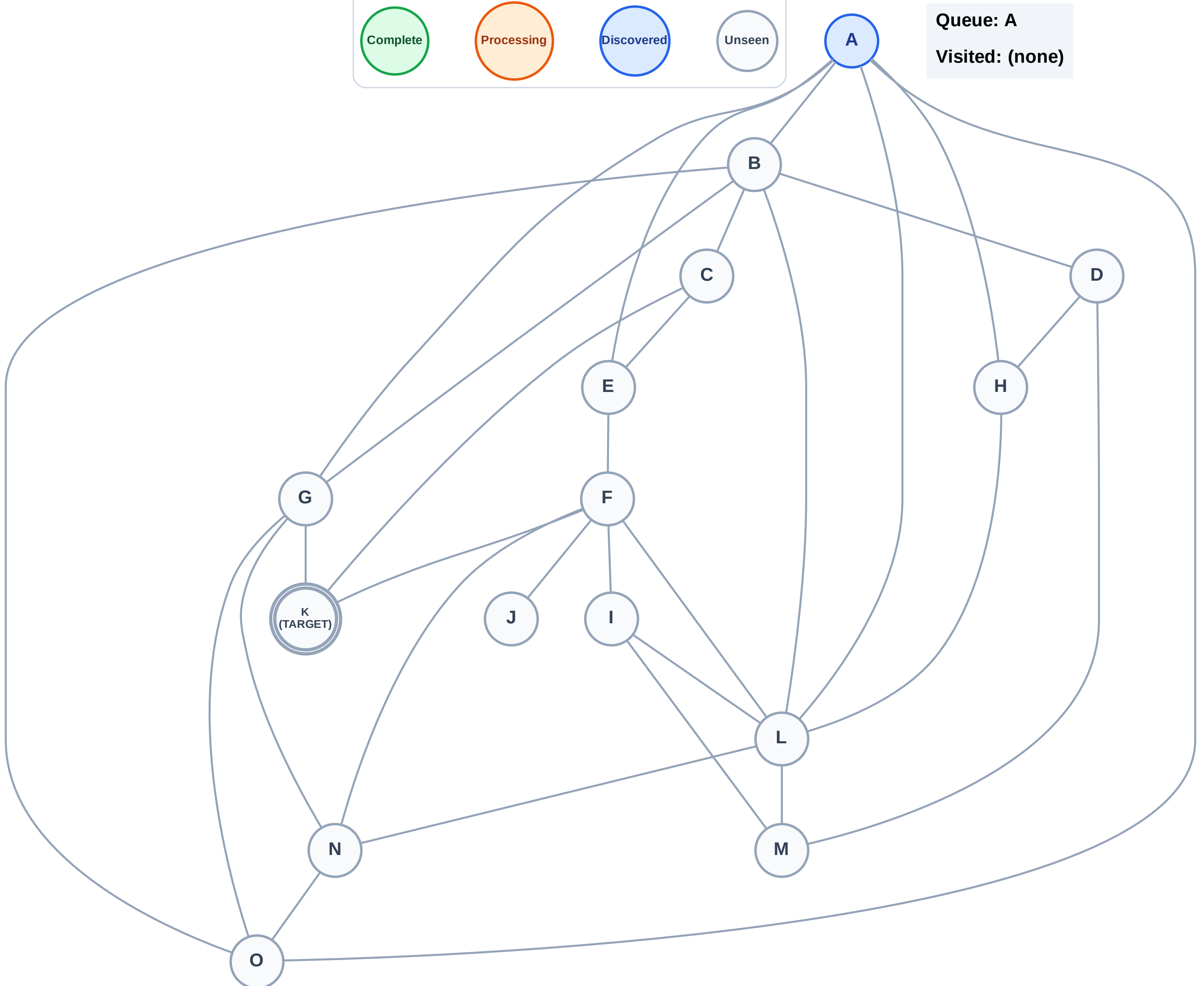
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

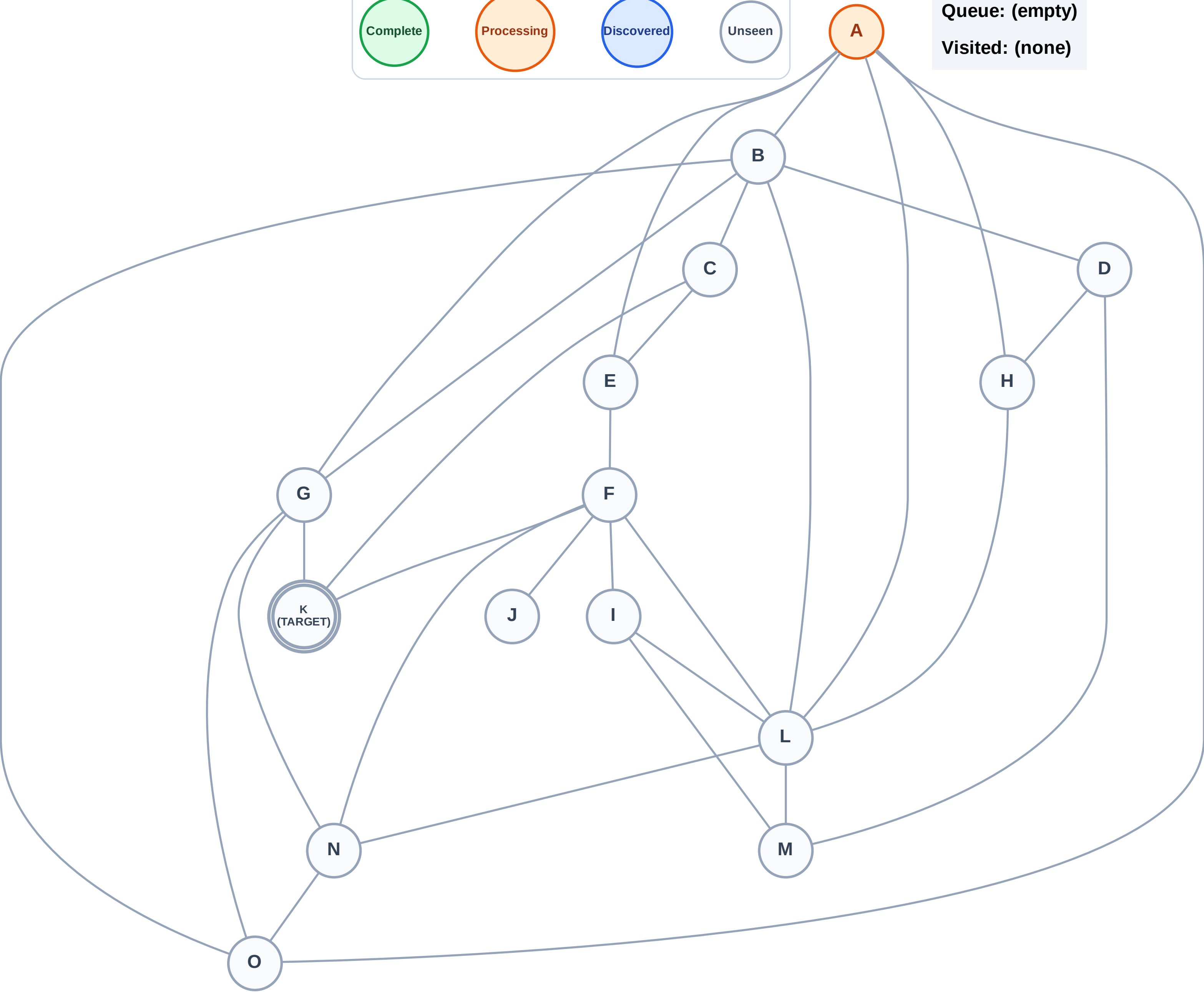
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



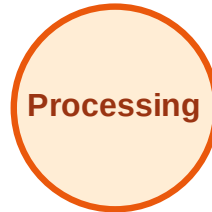
BFS Traversal

Step 3: Discover B, E, G, H, L, O from A

Legend



Complete



Processing



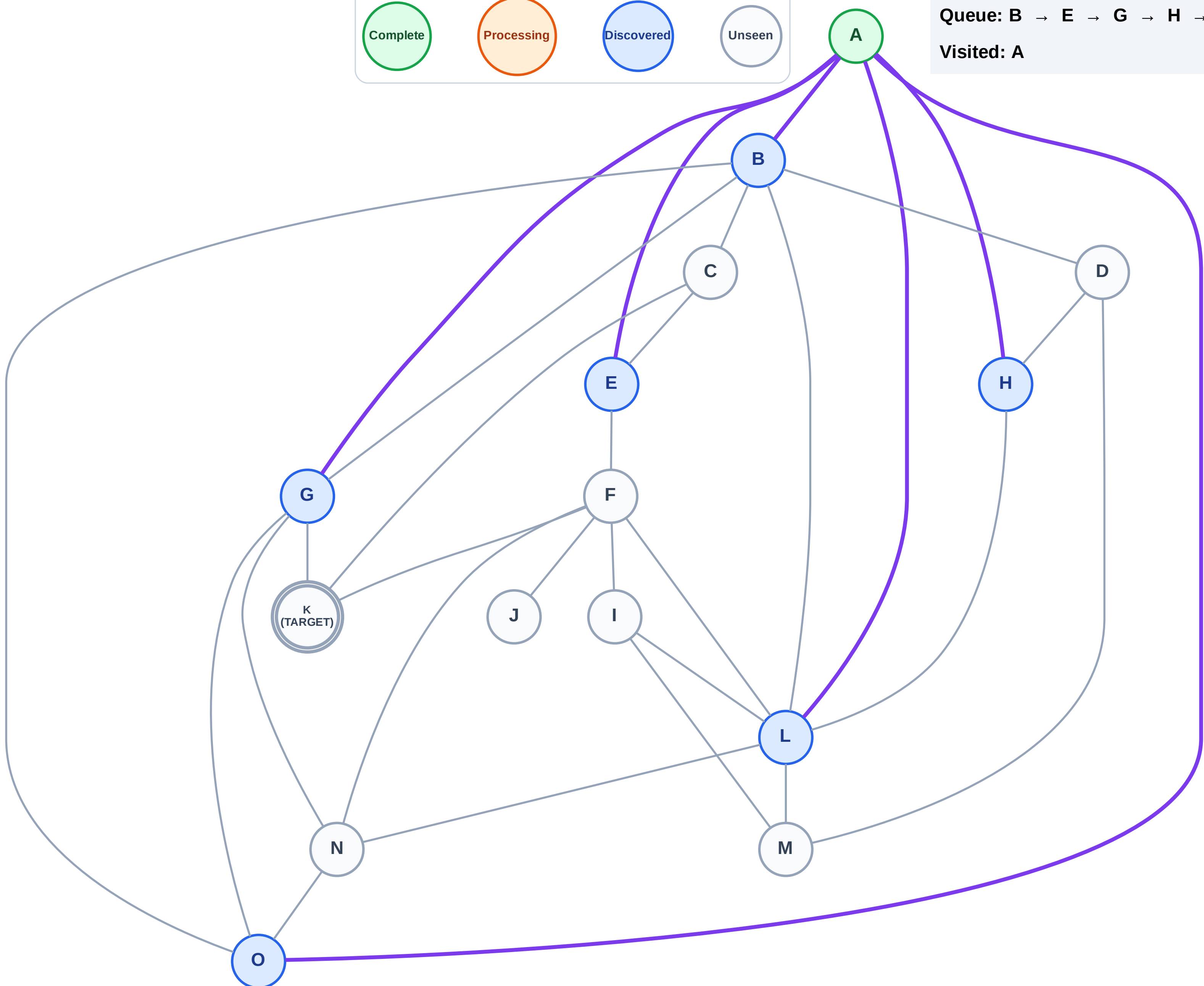
Discovered



Unseen

Queue: B → E → G → H → L → O

Visited: A



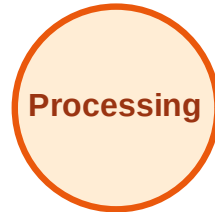
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



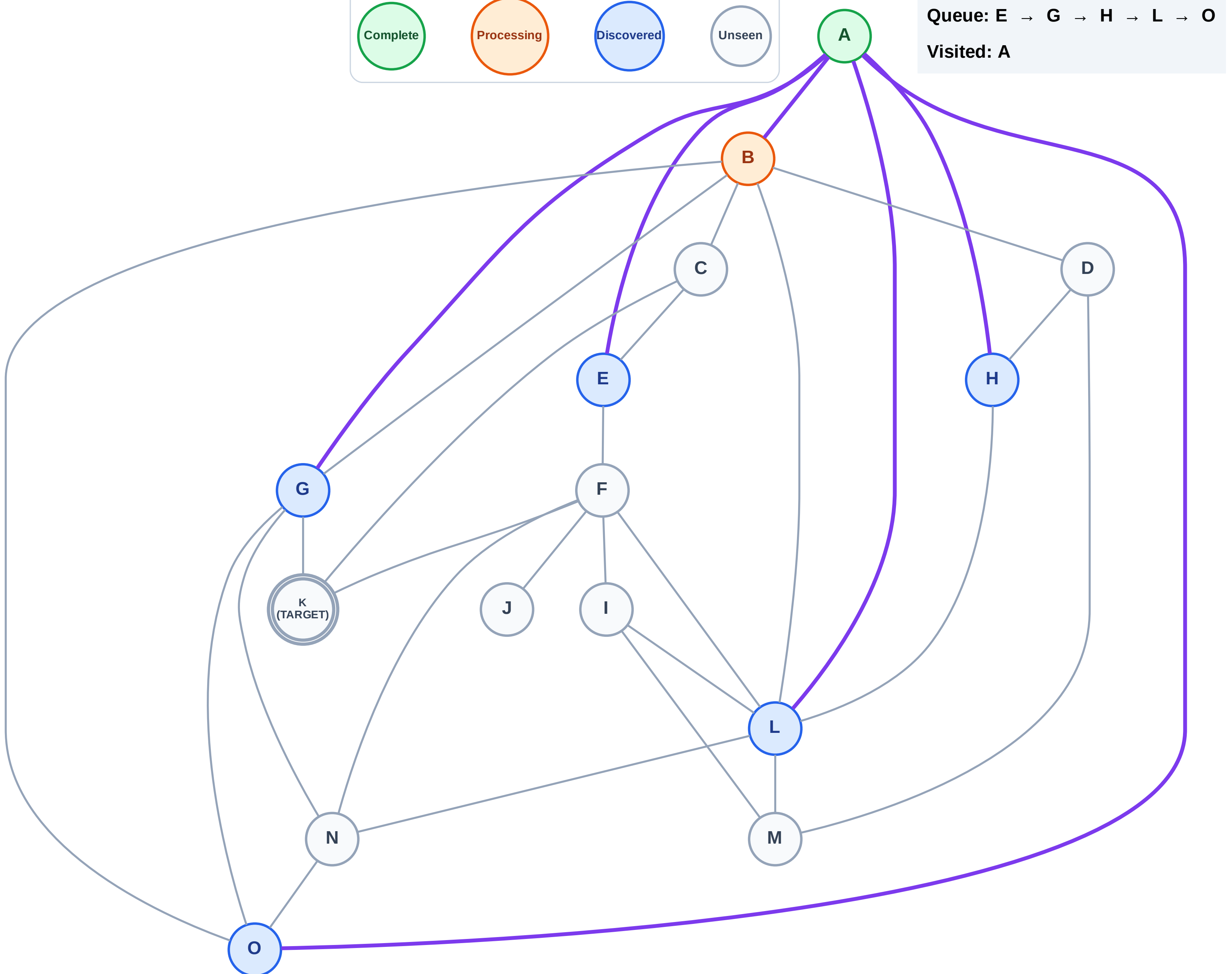
Discovered



Unseen

Queue: E → G → H → L → O

Visited: A



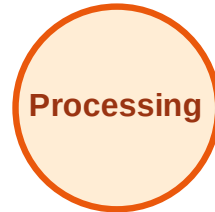
BFS Traversal

Step 5: Discover C, D from B

Legend



Complete



Processing



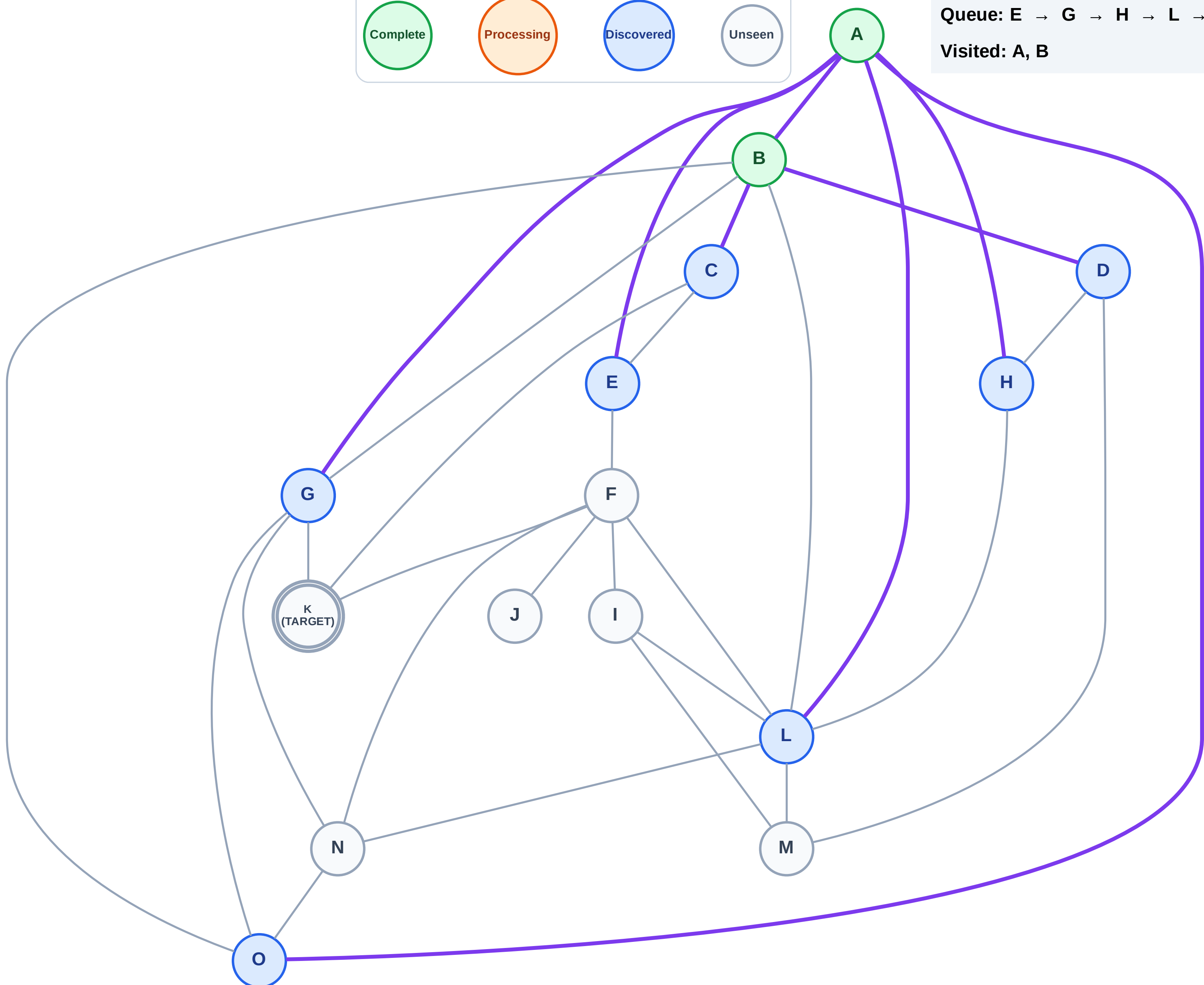
Discovered



Unseen

Queue: E → G → H → L → O → C → D

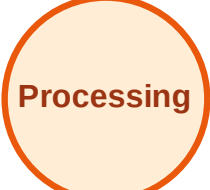
Visited: A, B



BFS Traversal

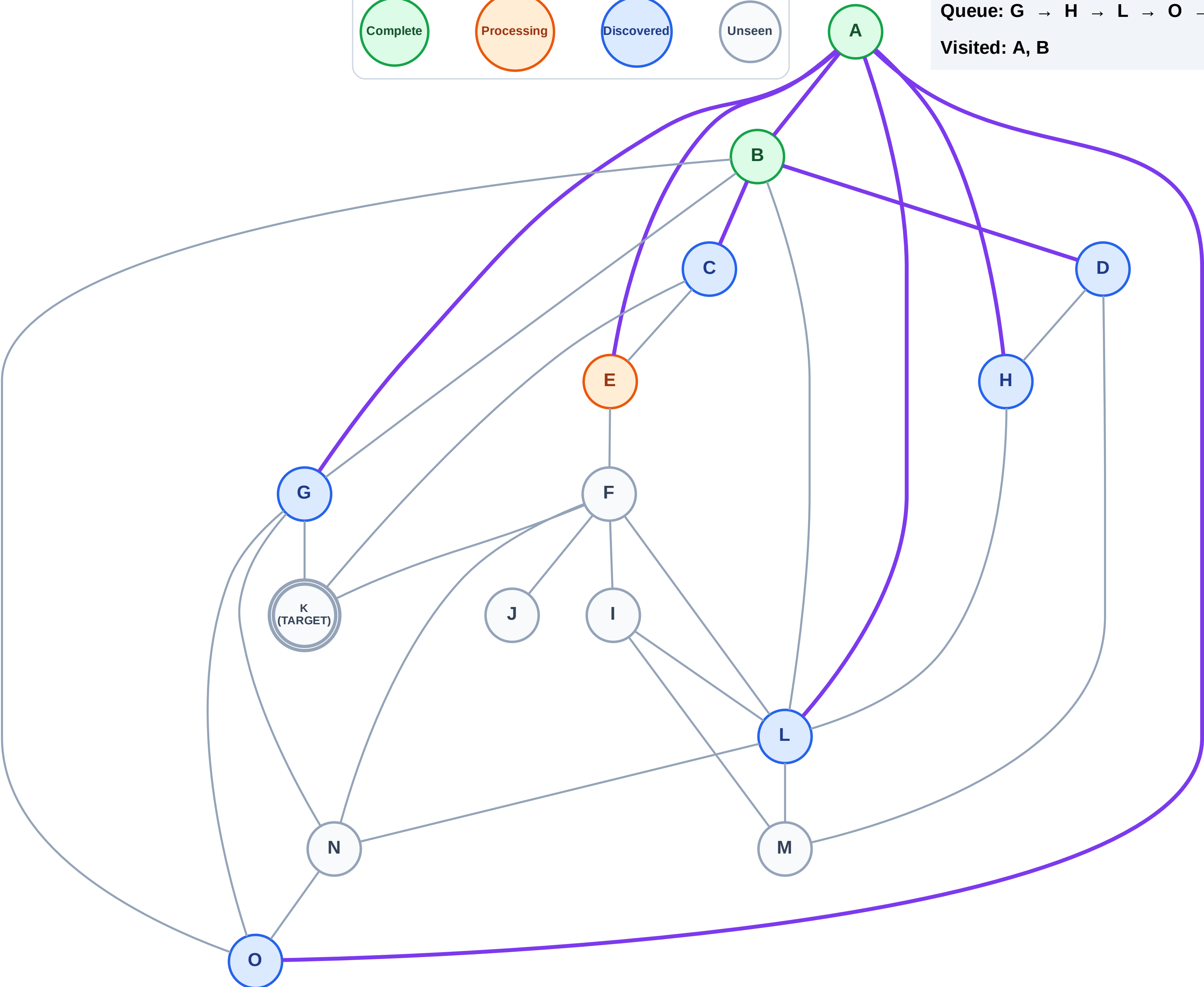
Step 6: Process node E

Legend



Queue: G → H → L → O → C → D

Visited: A, B



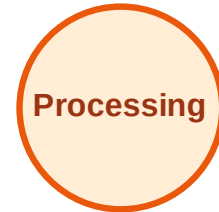
BFS Traversal

Step 7: Discover F from E

Legend



Complete



Processing



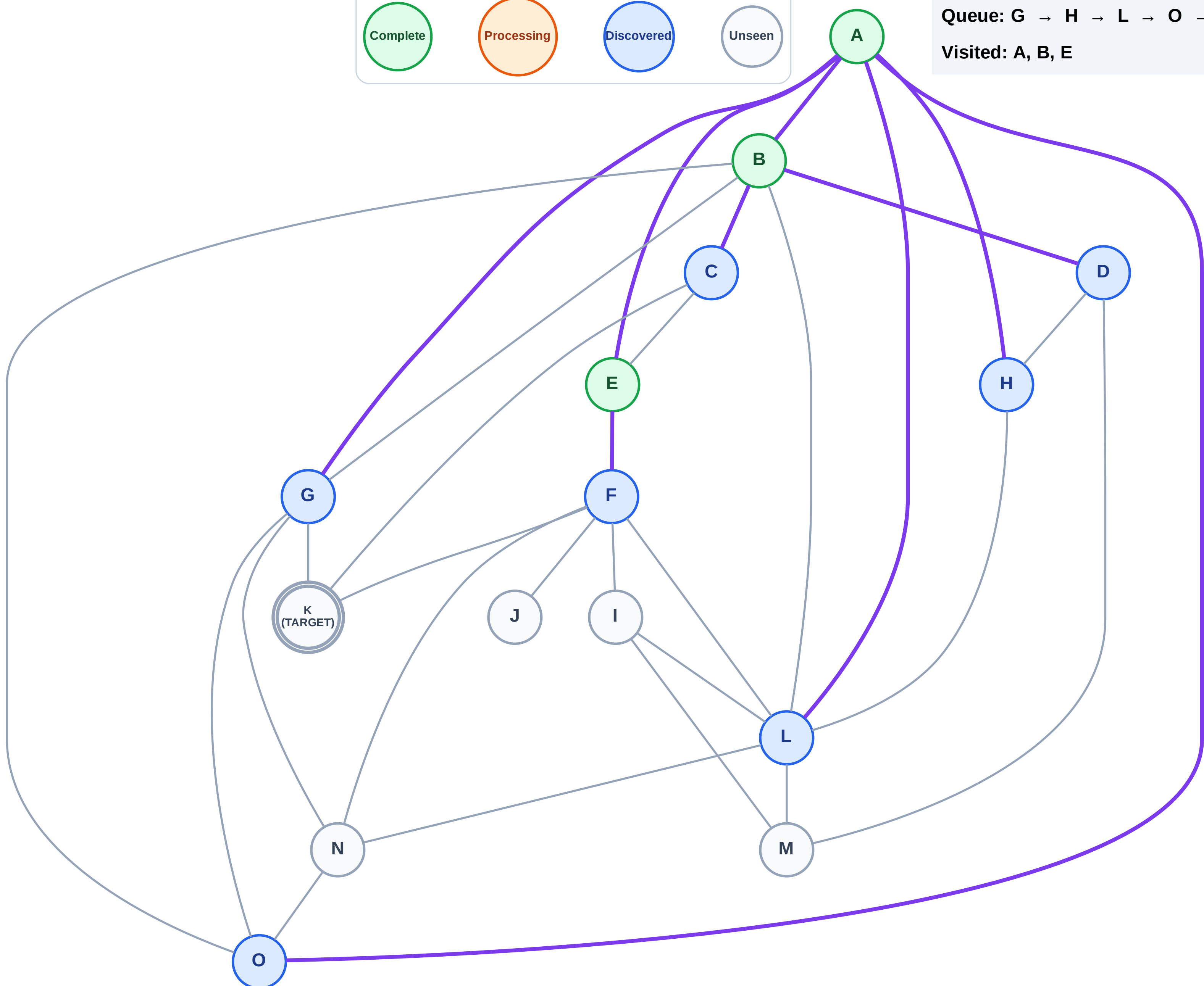
Discovered



Unseen

Queue: G → H → L → O → C → D → F

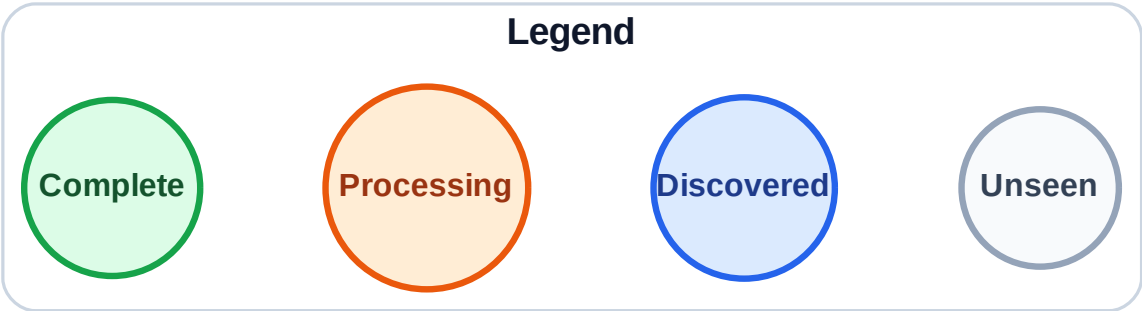
Visited: A, B, E



BFS Traversal

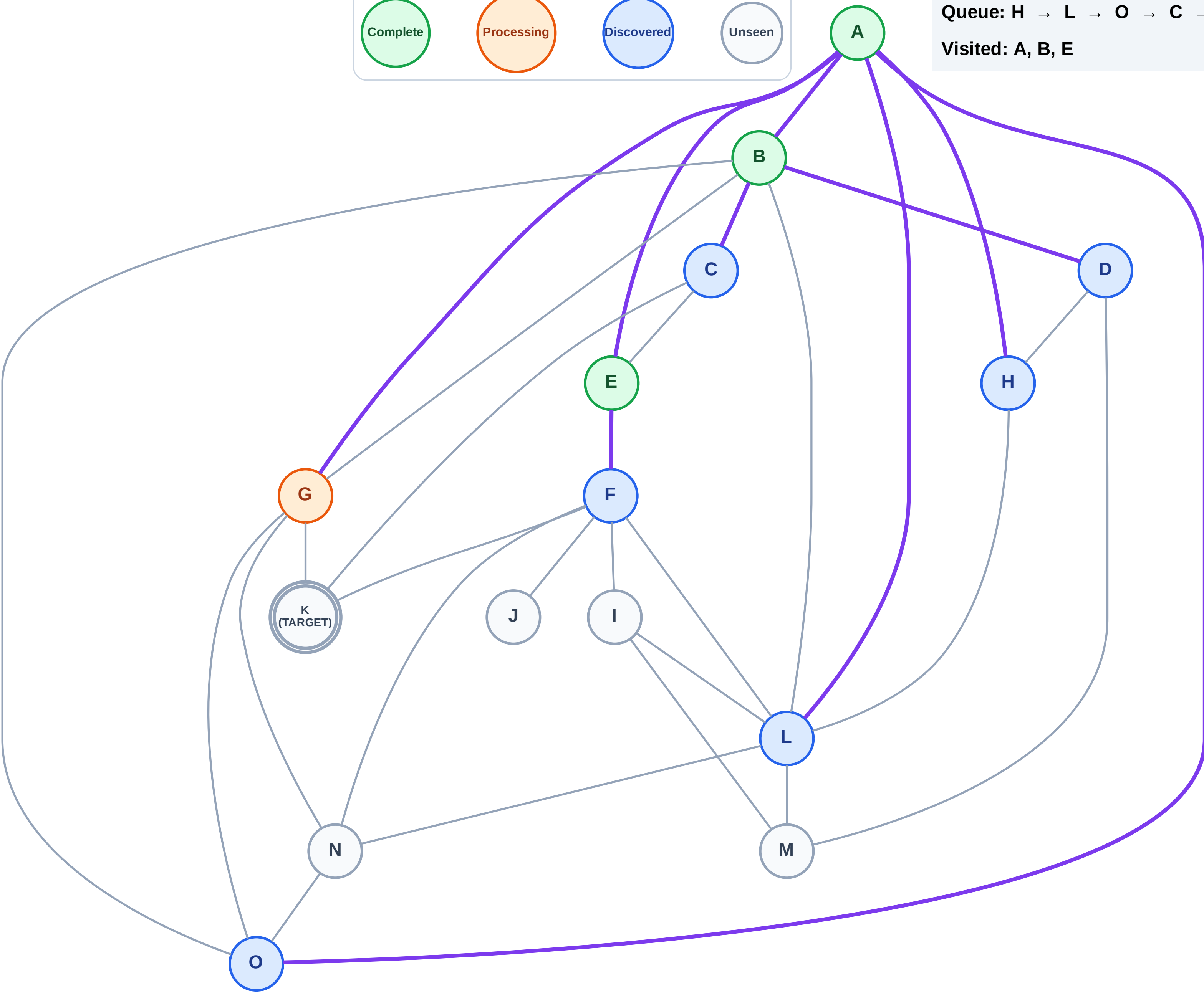
Step 8: Process node G

Legend



Queue: H → L → O → C → D → F

Visited: A, B, E



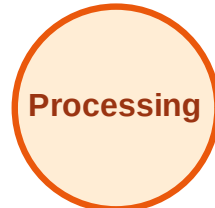
BFS Traversal

Step 9: Discover K, N from G

Legend



Complete



Processing



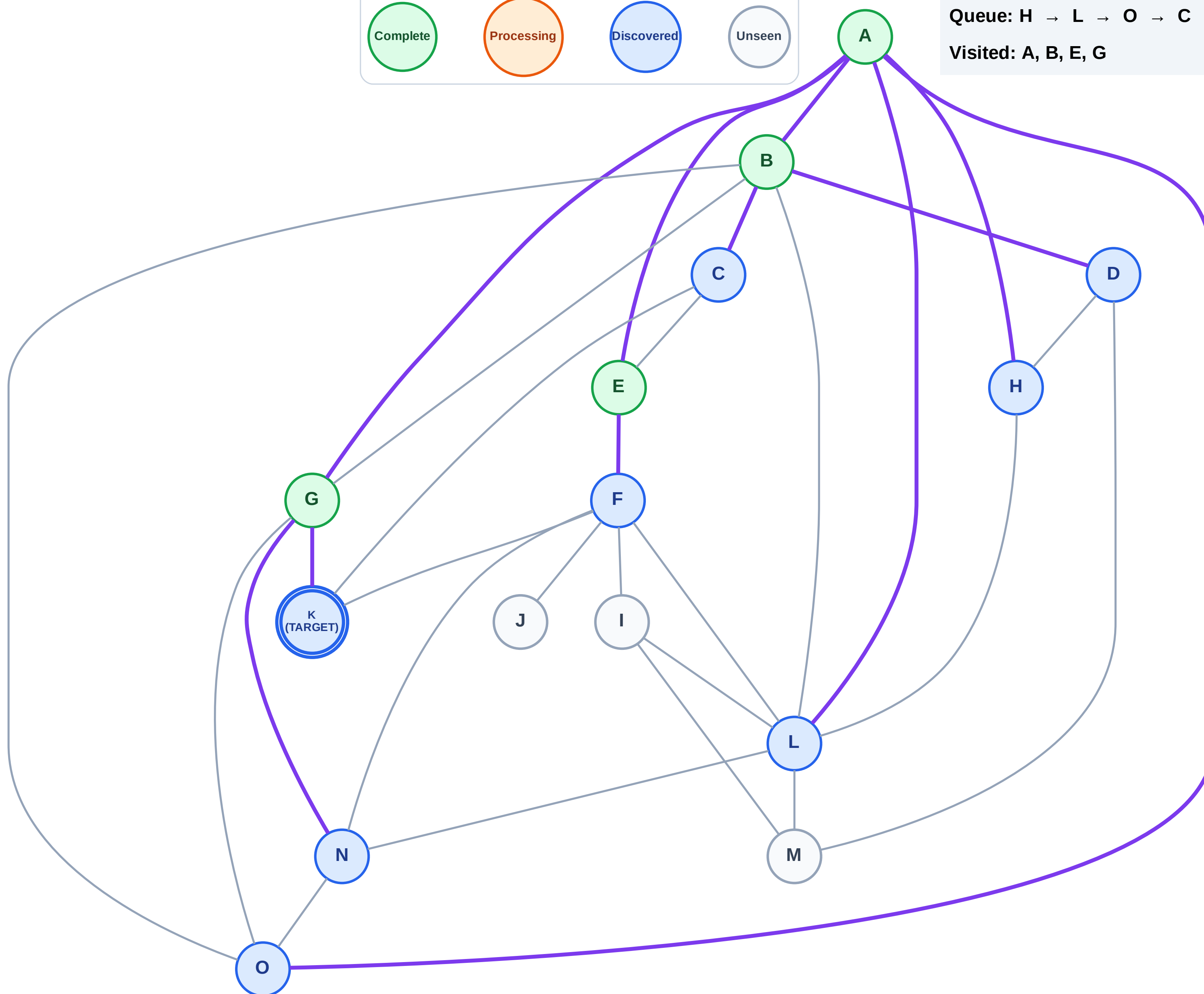
Discovered



Unseen

Queue: H → L → O → C → D → F → K → N

Visited: A, B, E, G



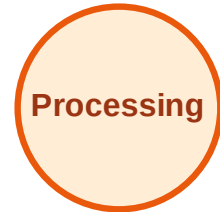
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



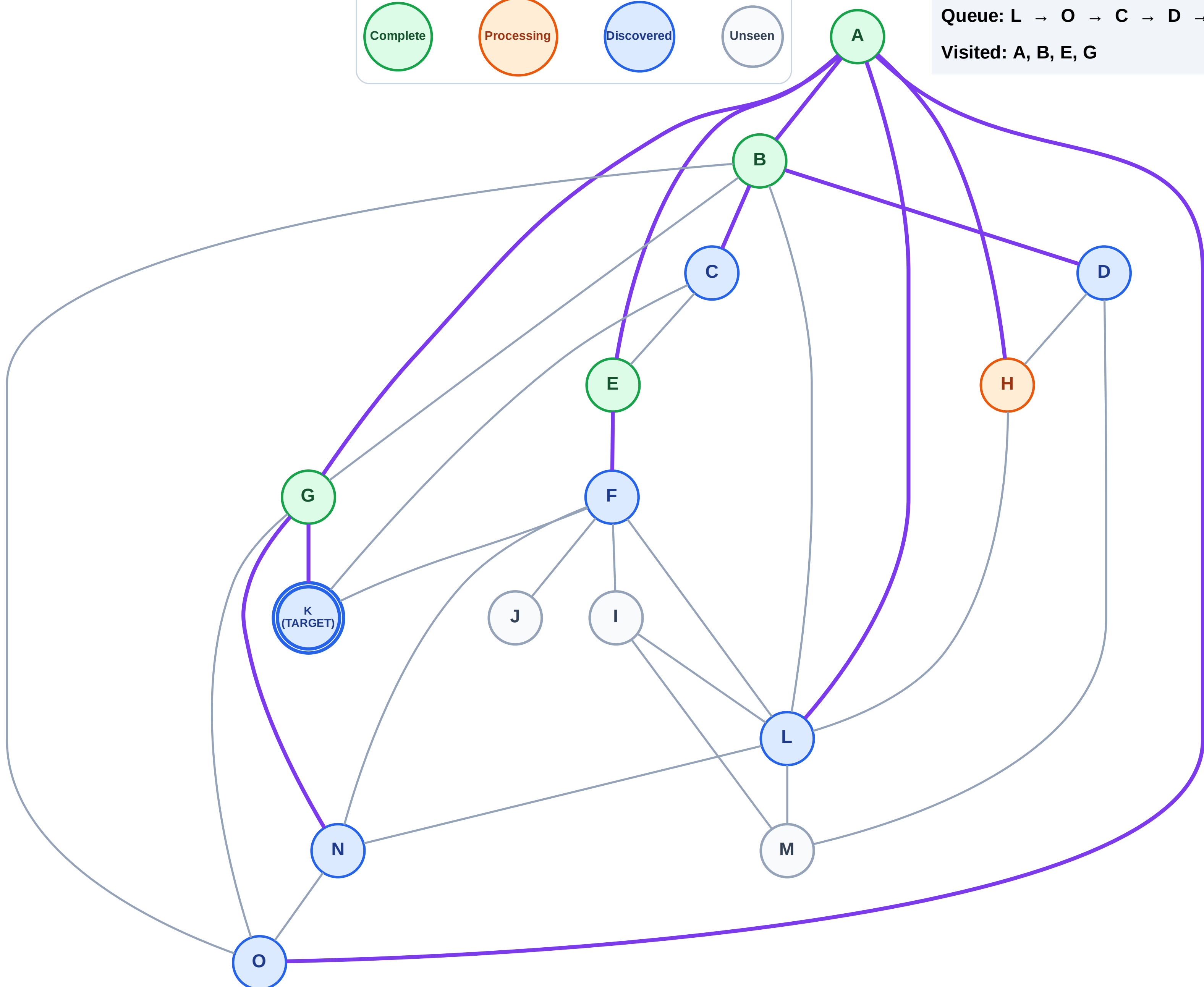
Discovered



Unseen

Queue: L → O → C → D → F → K → N

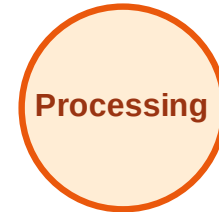
Visited: A, B, E, G



BFS Traversal

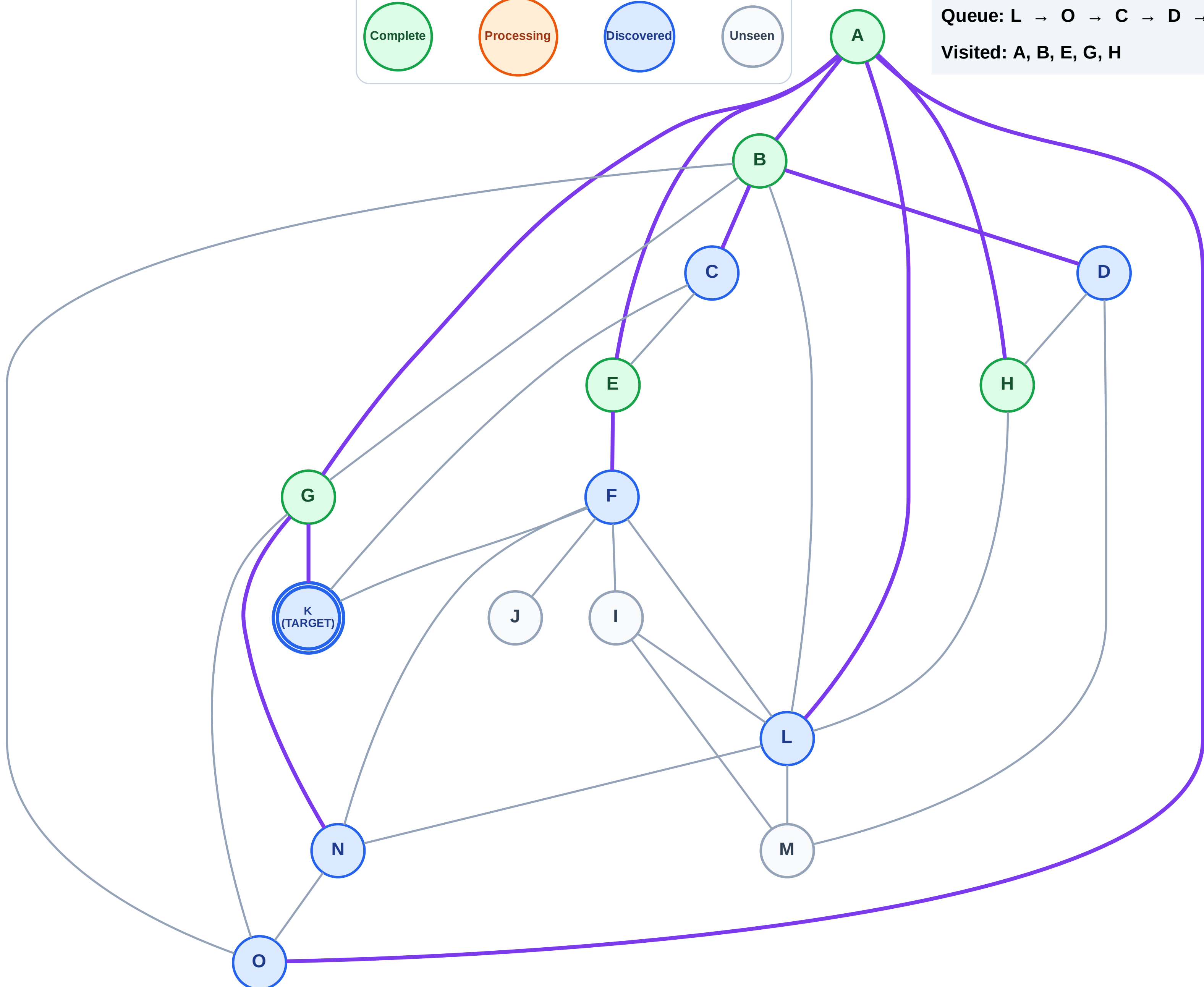
Step 11: No new neighbors from H

Legend



Queue: L → O → C → D → F → K → N

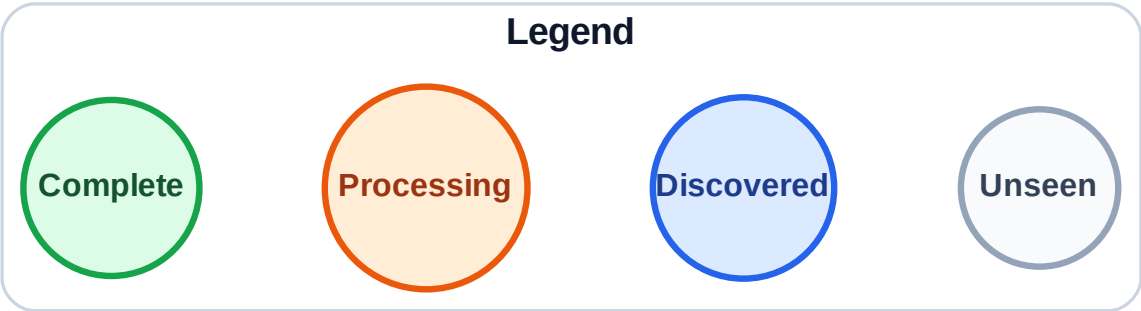
Visited: A, B, E, G, H



BFS Traversal

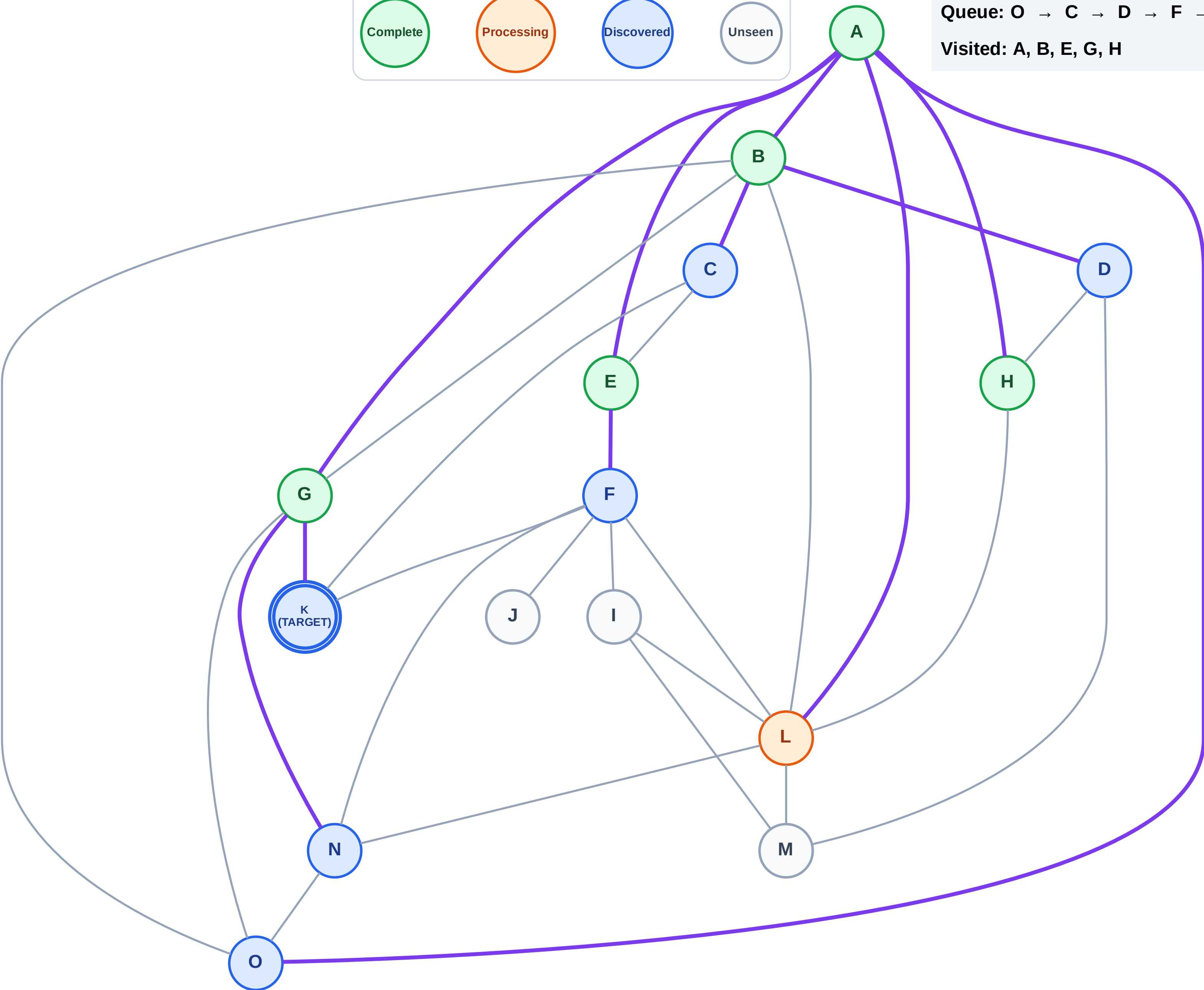
Step 12: Process node L

Legend



Queue: O → C → D → F → K → N

Visited: A, B, E, G, H



BFS Traversal

Step 13: Discover I, M from L

Legend



Complete



Processing



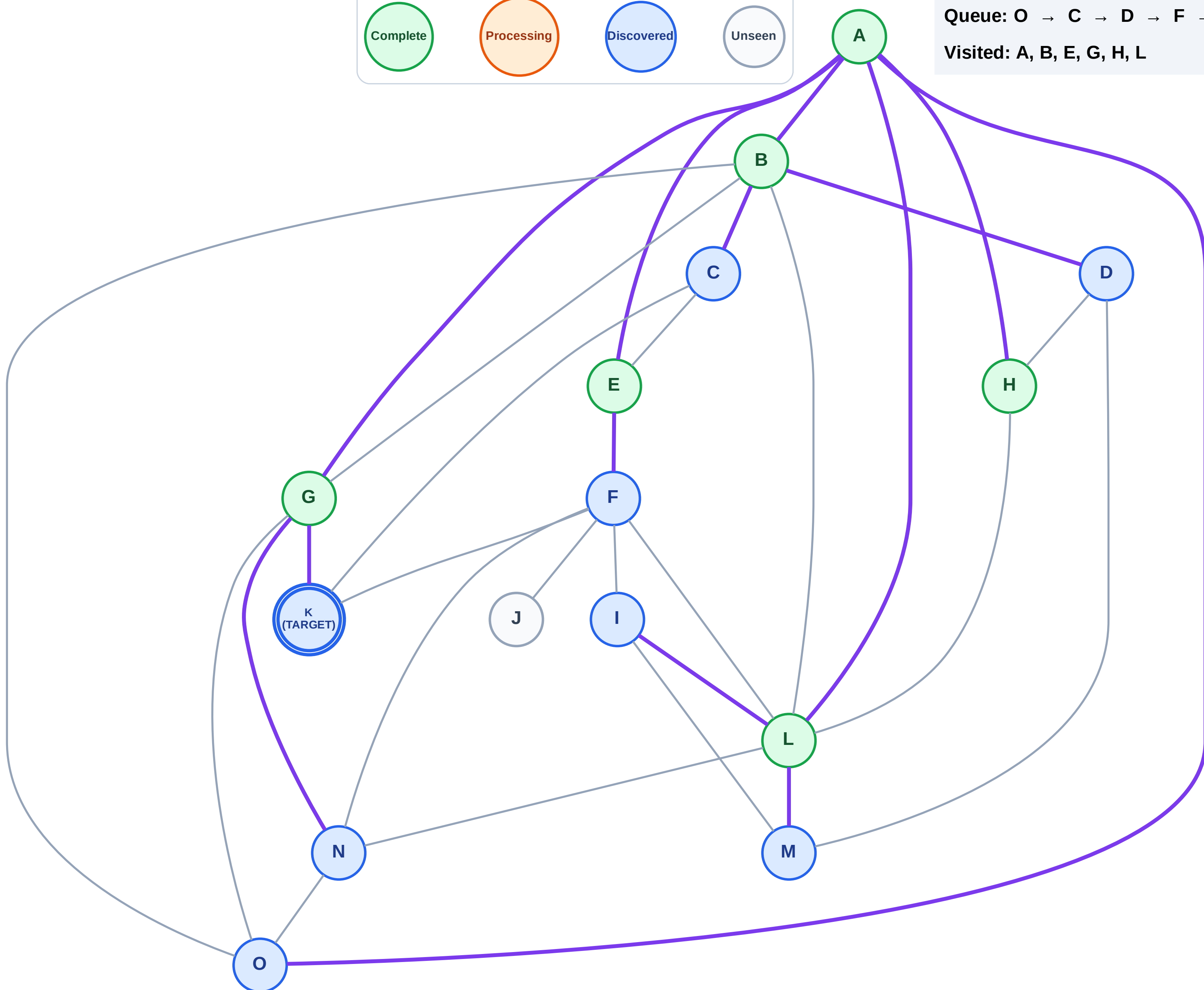
Discovered



Unseen

Queue: O → C → D → F → K → N → I → M

Visited: A, B, E, G, H, L



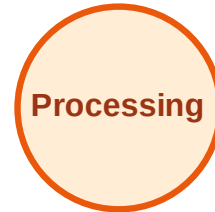
BFS Traversal

Step 14: Process node O

Legend



Complete



Processing



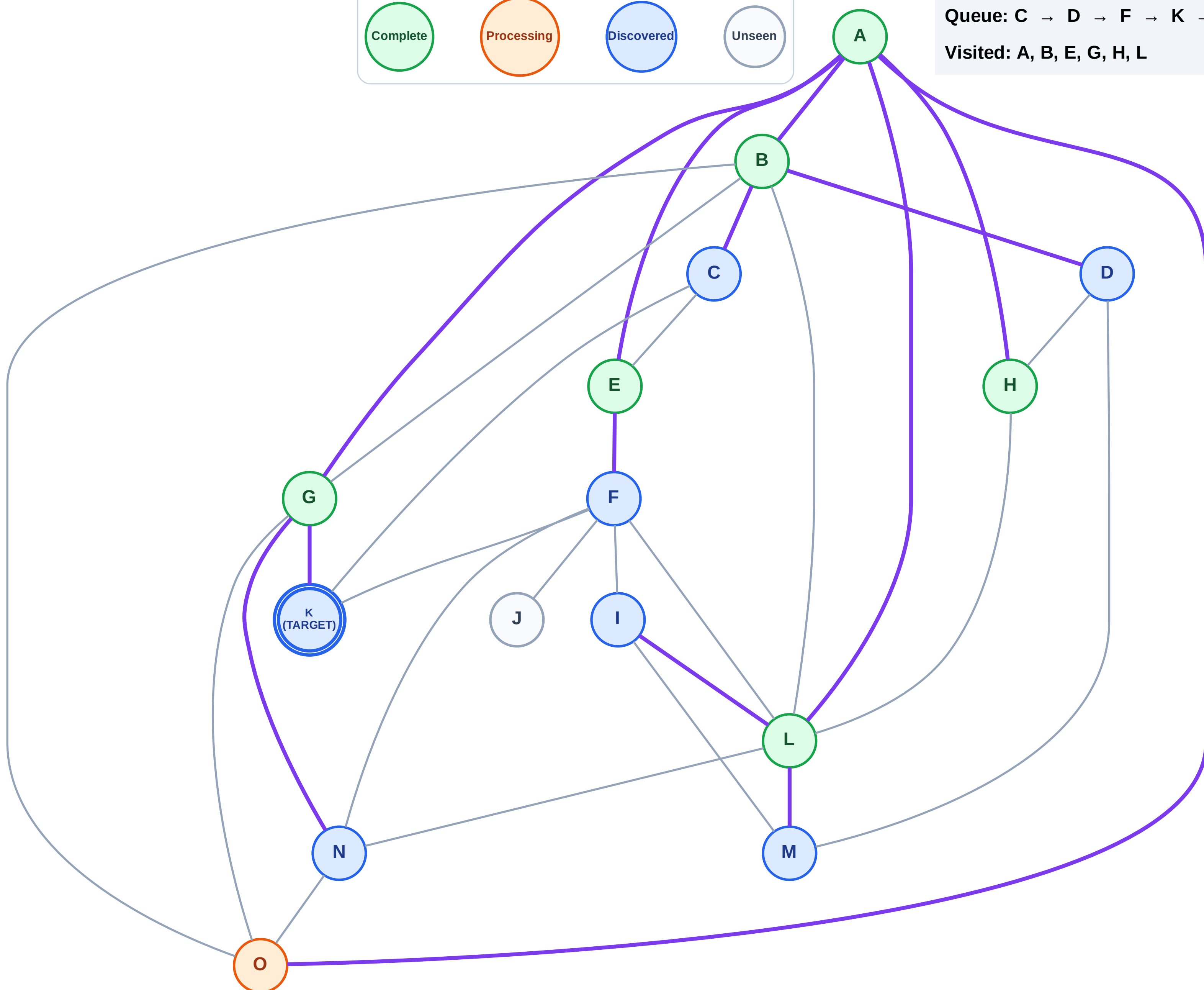
Discovered



Unseen

Queue: C → D → F → K → N → I → M

Visited: A, B, E, G, H, L



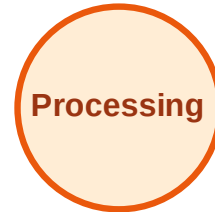
BFS Traversal

Step 15: No new neighbors from O

Legend



Complete



Processing



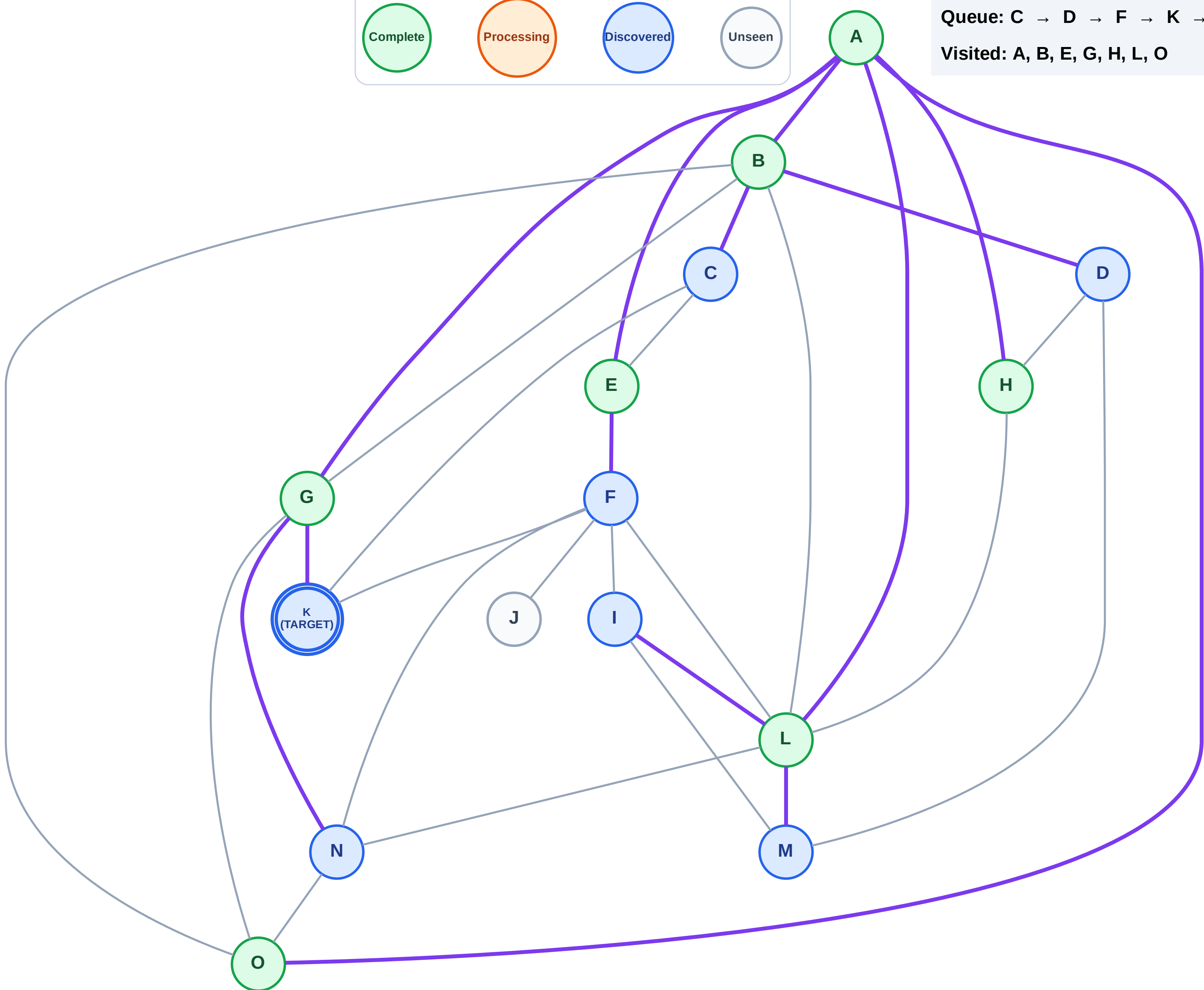
Discovered



Unseen

Queue: C → D → F → K → N → I → M

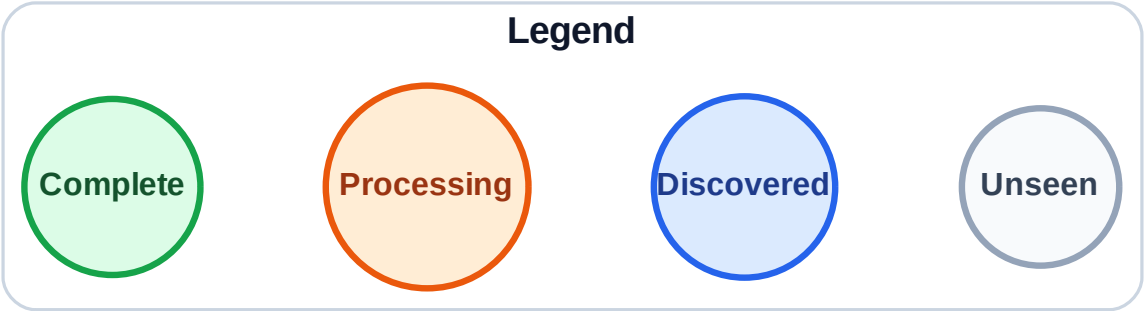
Visited: A, B, E, G, H, L, O



BFS Traversal

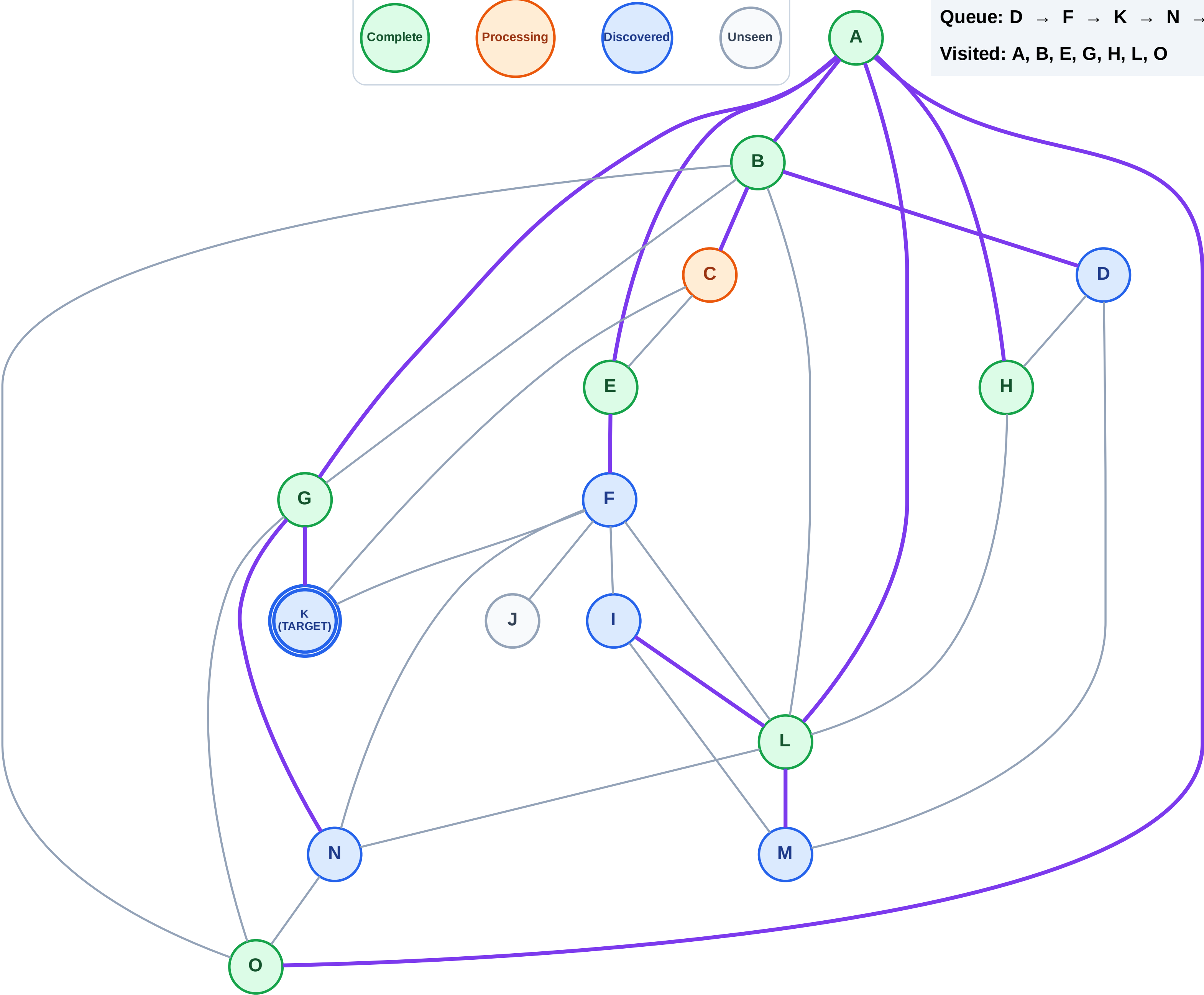
Step 16: Process node C

Legend



Queue: D → F → K → N → I → M

Visited: A, B, E, G, H, L, O



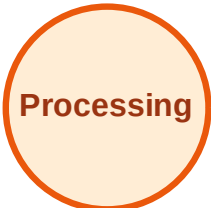
BFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



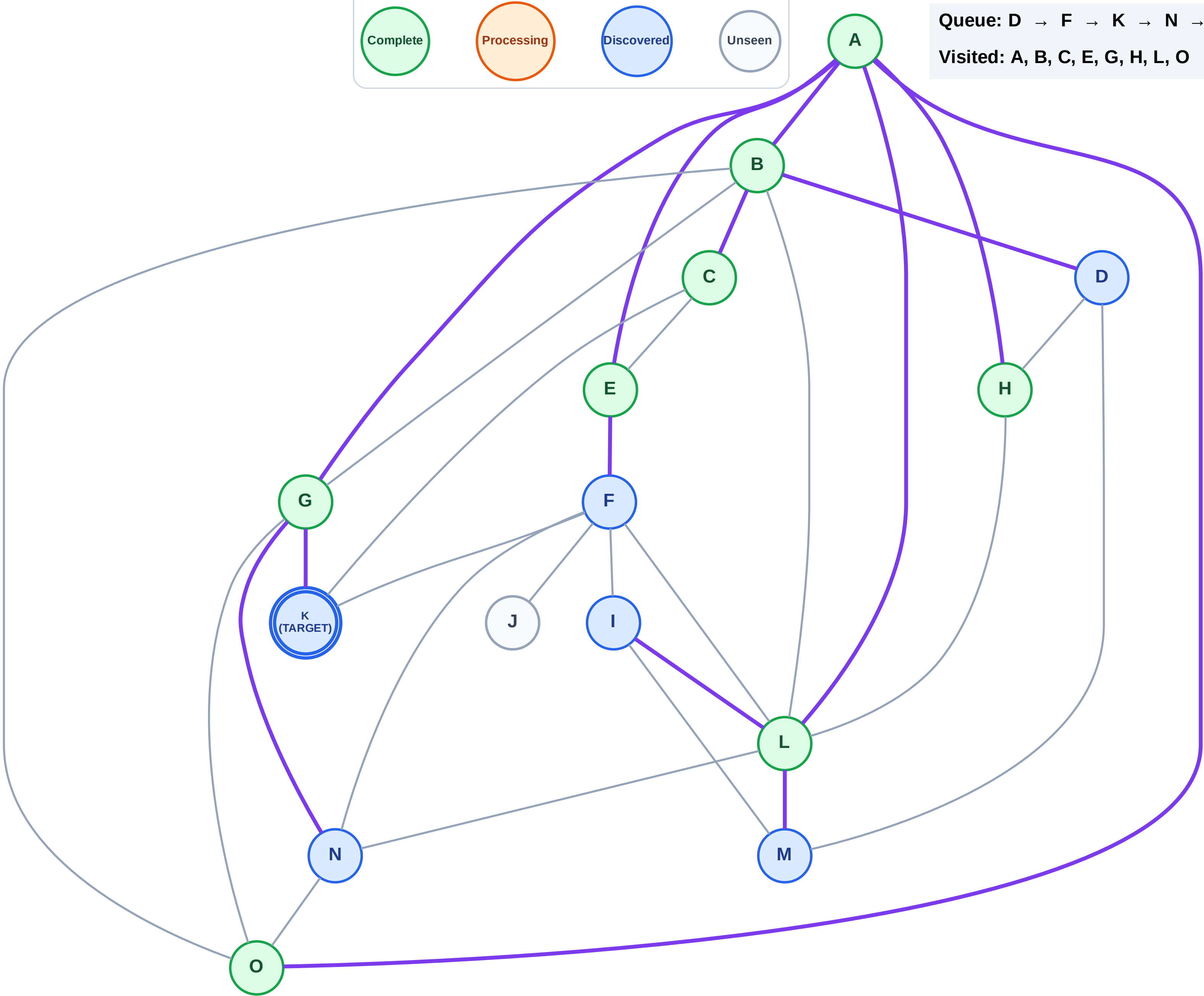
Discovered



Unseen

Queue: D → F → K → N → I → M

Visited: A, B, C, E, G, H, L, O



BFS Traversal

Step 18: Process node D

Legend

Complete

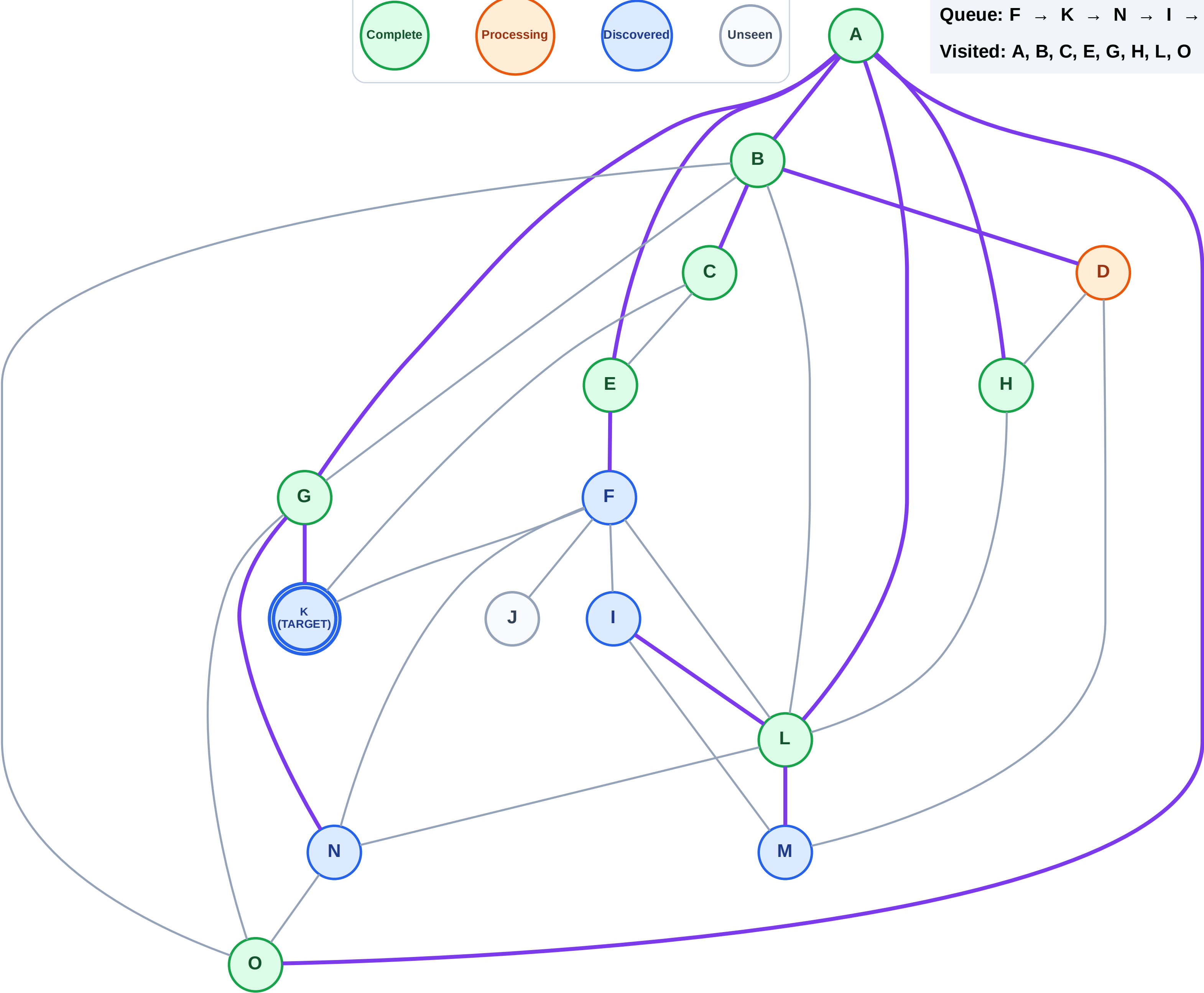
Processing

Discovered

Unseen

Queue: F → K → N → I → M

Visited: A, B, C, E, G, H, L, O



BFS Traversal

Step 19: No new neighbors from D

Legend



Complete



Processing



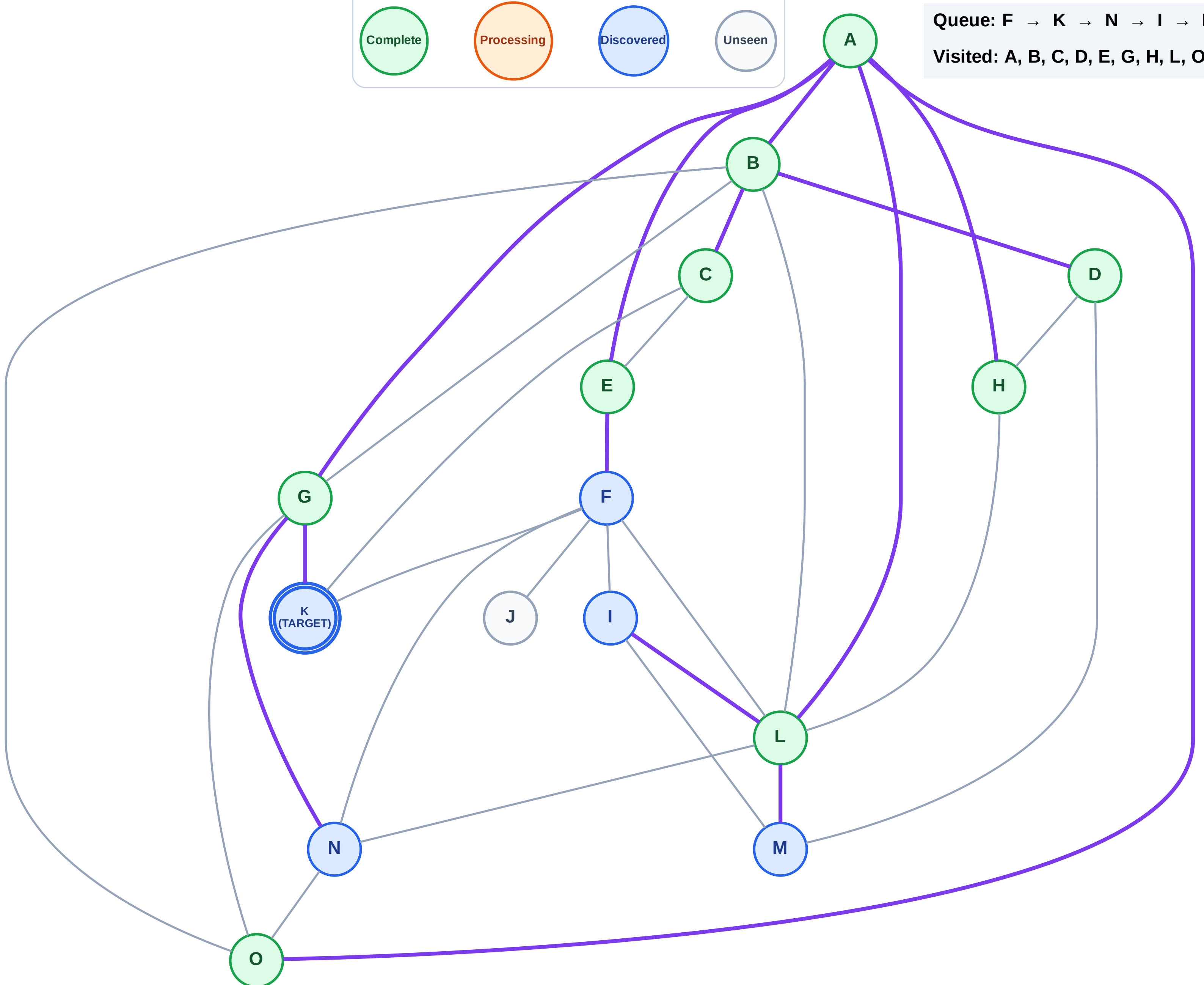
Discovered



Unseen

Queue: F → K → N → I → M

Visited: A, B, C, D, E, G, H, L, O



BFS Traversal

Step 20: Process node F

Legend

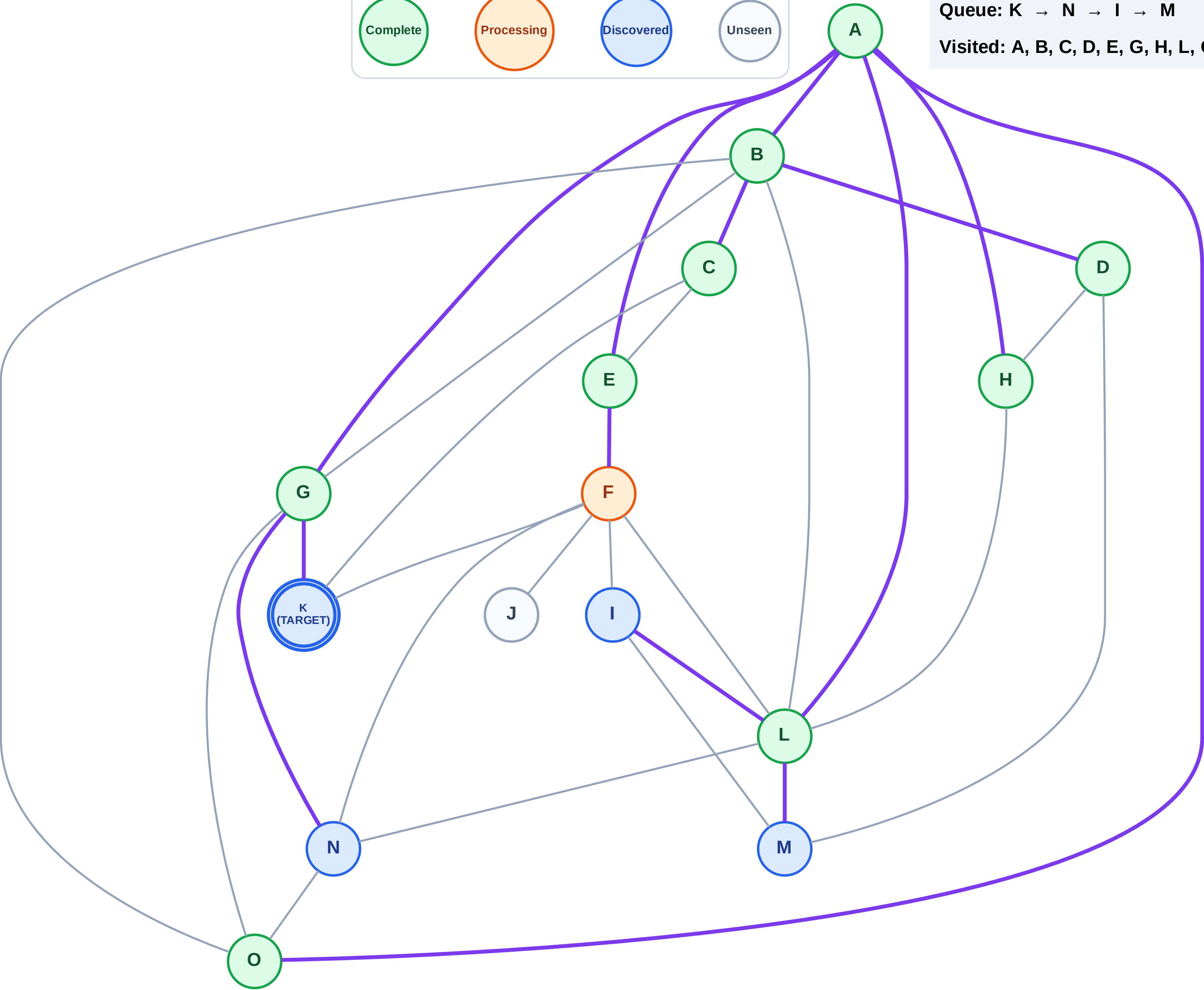
Complete

Processing

Discovered

Unseen

Queue: K → N → I → M
Visited: A, B, C, D, E, G, H, L, O



Step 21: Discover J from F

Step 21: Discover J from F

Legend

Complete

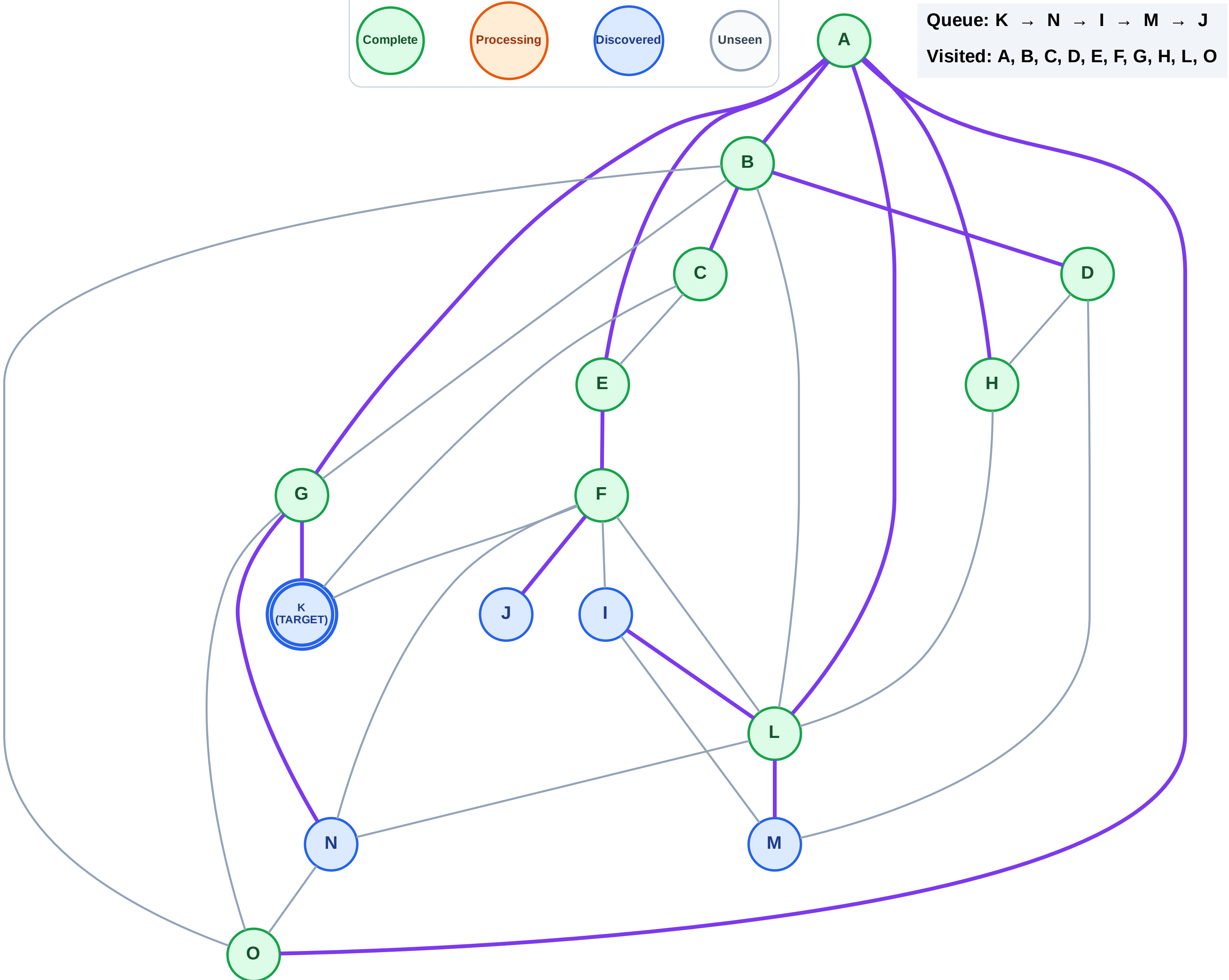
Processing

Discovered

Unseen

Queue: K → N → I → M → J

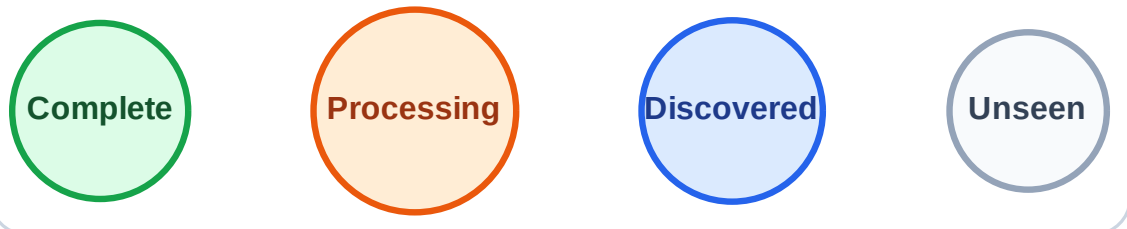
Visited: A, B, C, D, E, F, G, H, L, O



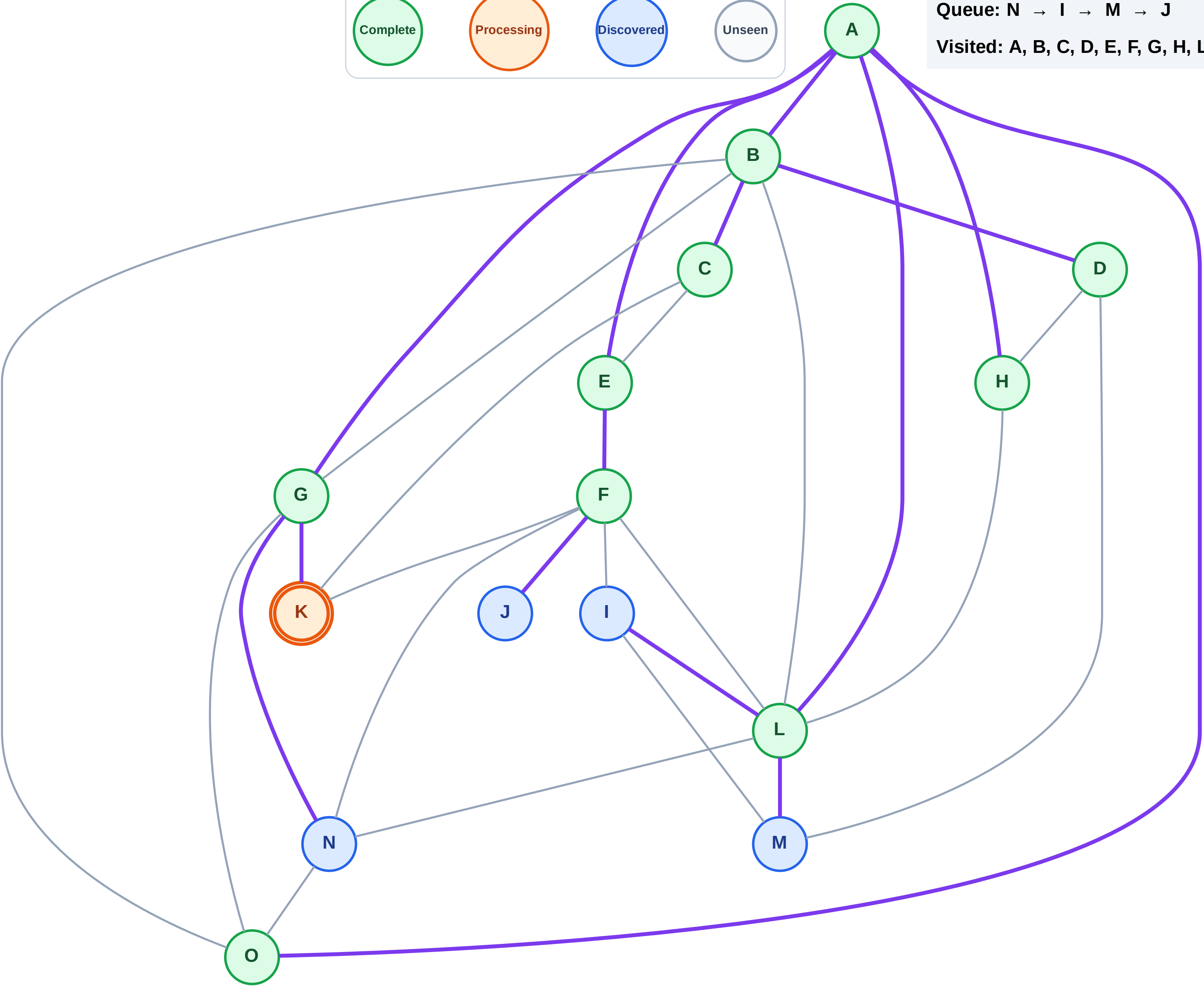
BFS Traversal

Step 22: Process node K

Legend



Queue: N → I → M → J
Visited: A, B, C, D, E, F, G, H, L, O



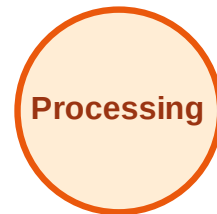
BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend



Complete



Processing



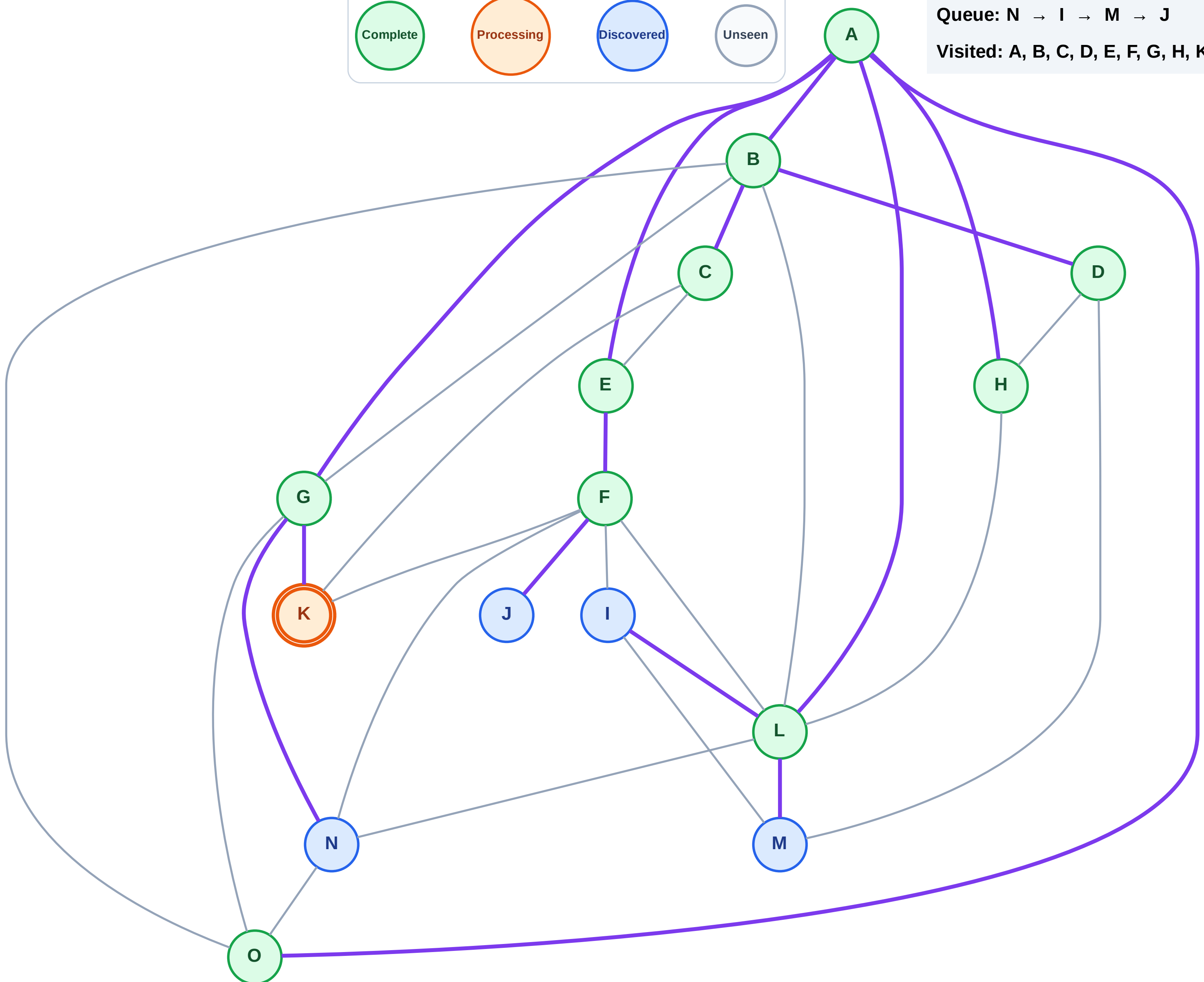
Discovered



Unseen

Queue: N → I → M → J

Visited: A, B, C, D, E, F, G, H, K, L, O



DFS Traversal

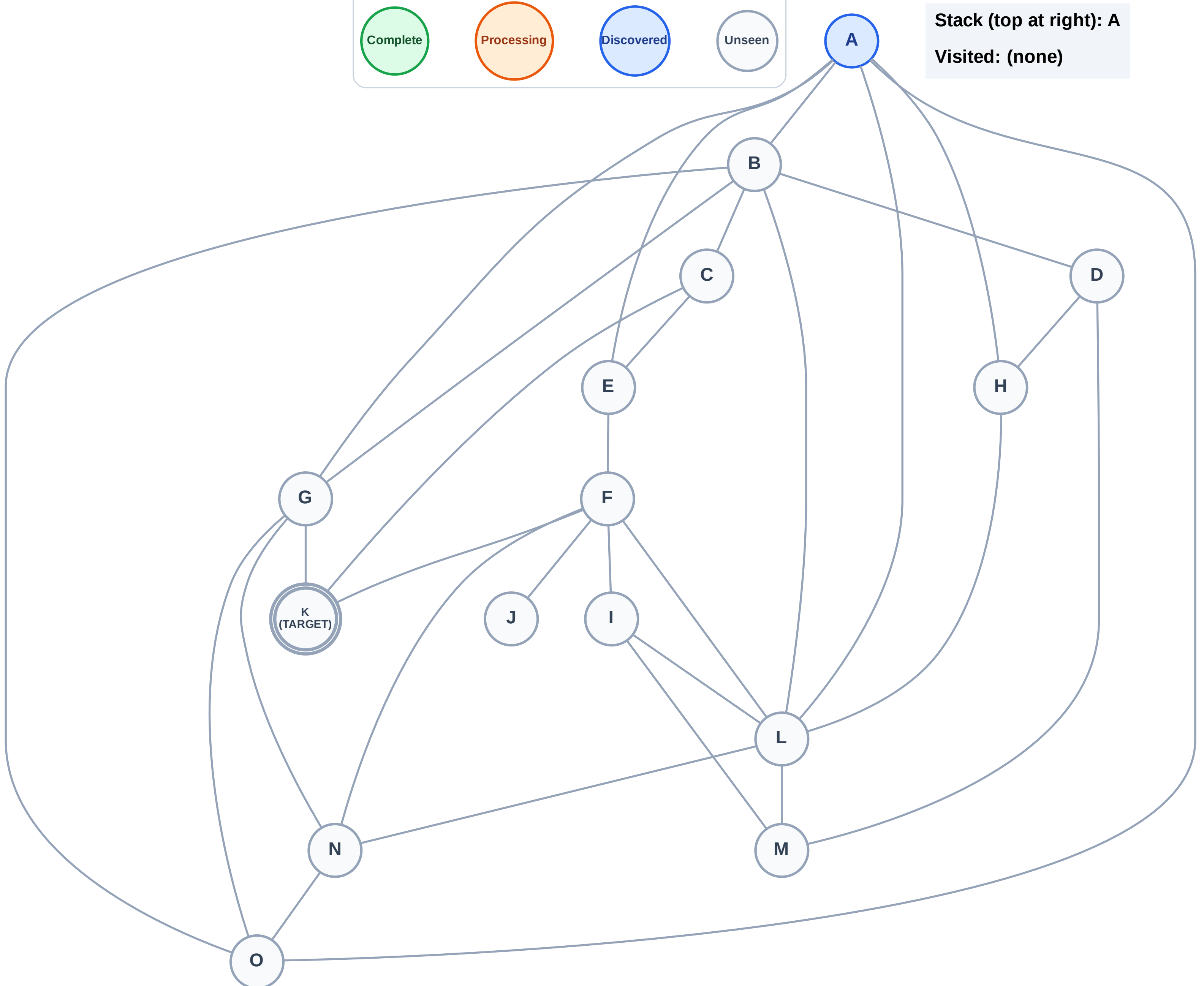
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

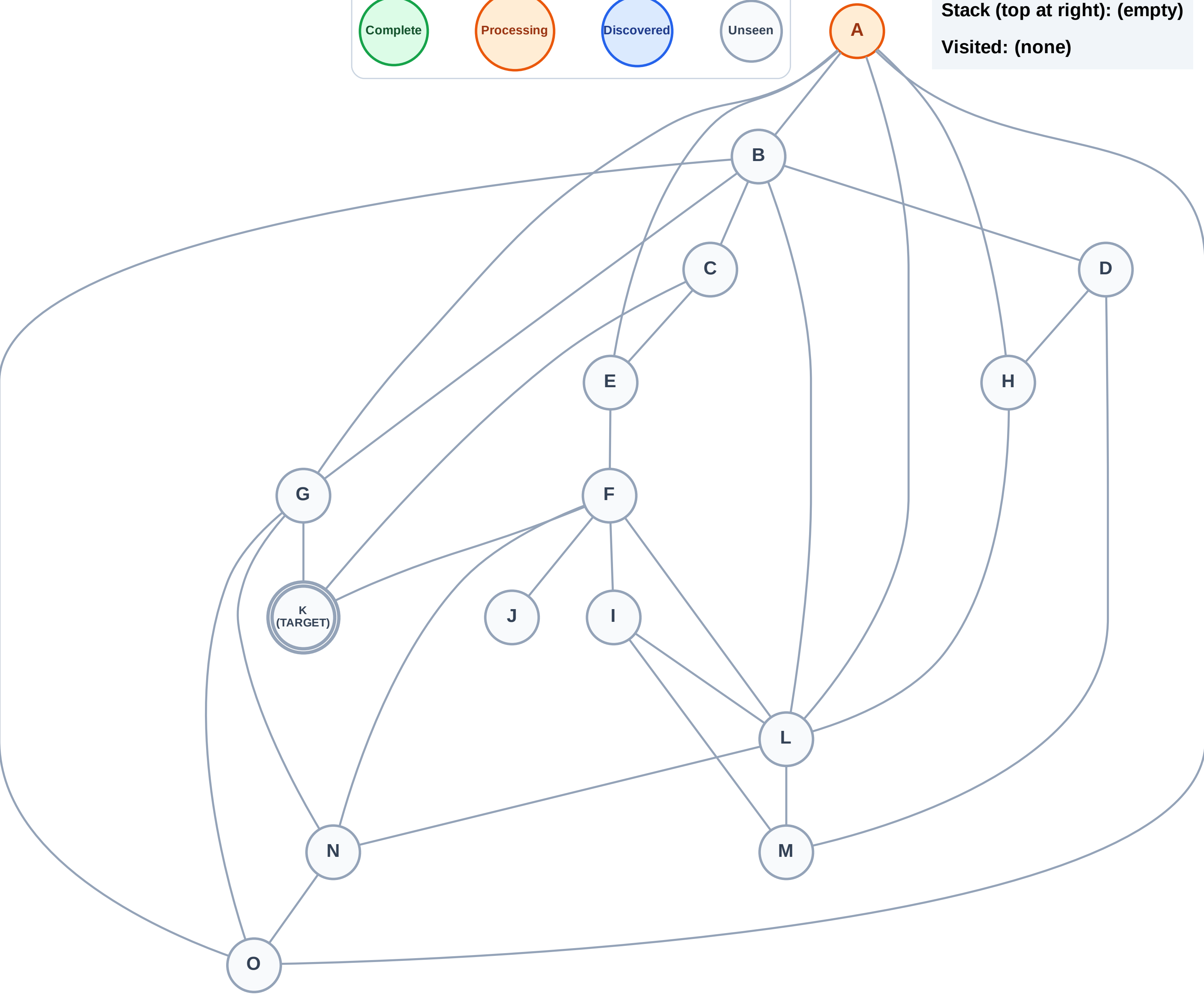
Complete

Processing

Discovered

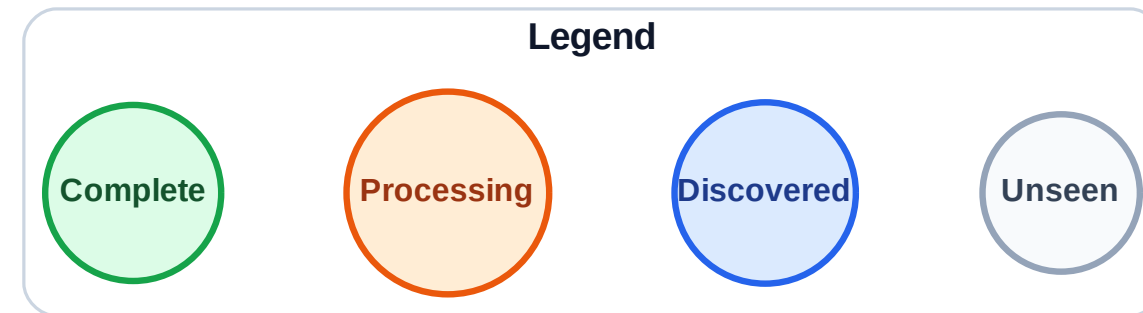
Unseen

Stack (top at right): (empty)
Visited: (none)



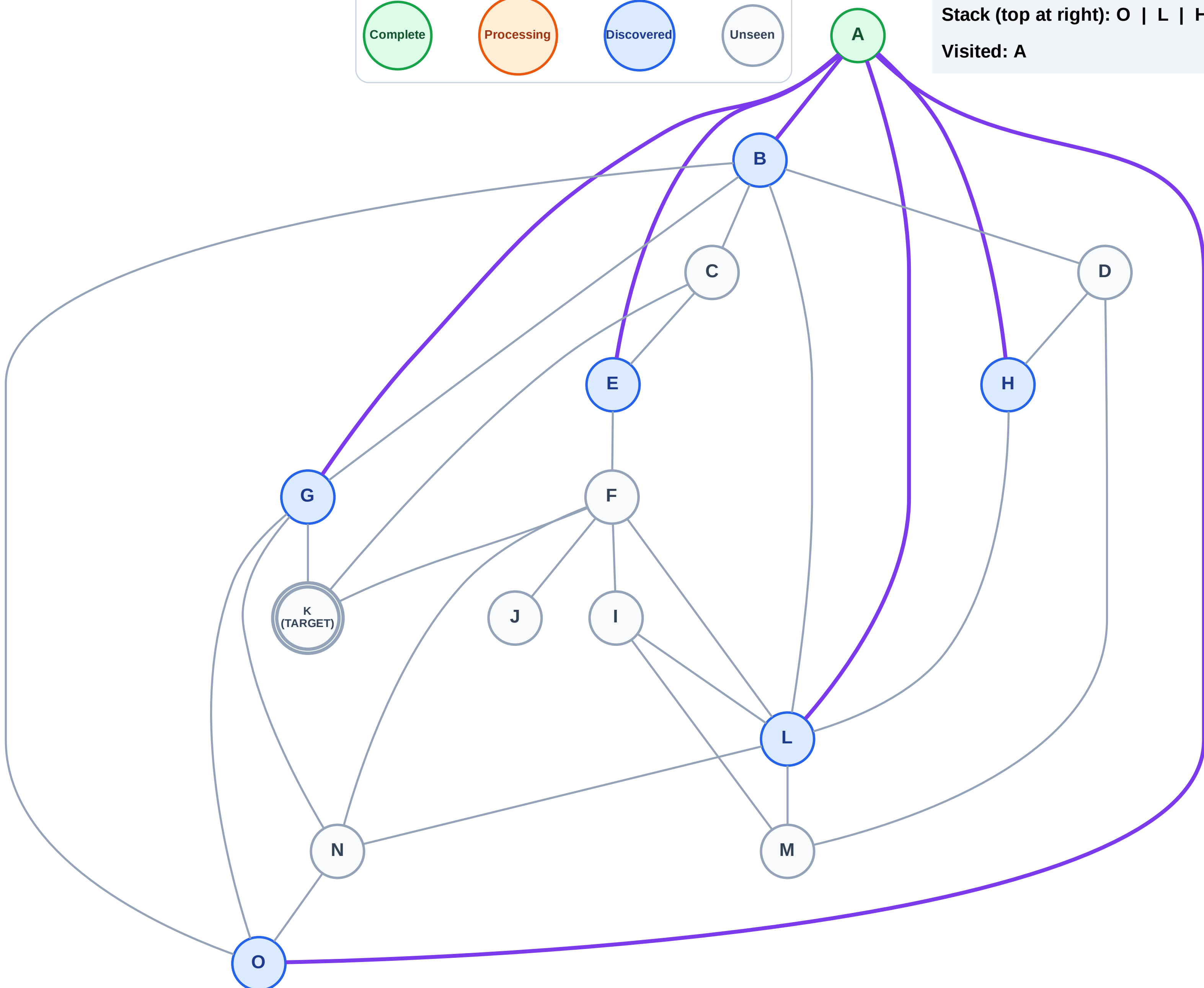
DFS Traversal

Step 3: Discover O, L, H, G, E, B from A



Stack (top at right): O | L | H | G | E | B

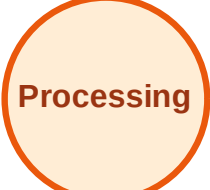
Visited: A



DFS Traversal

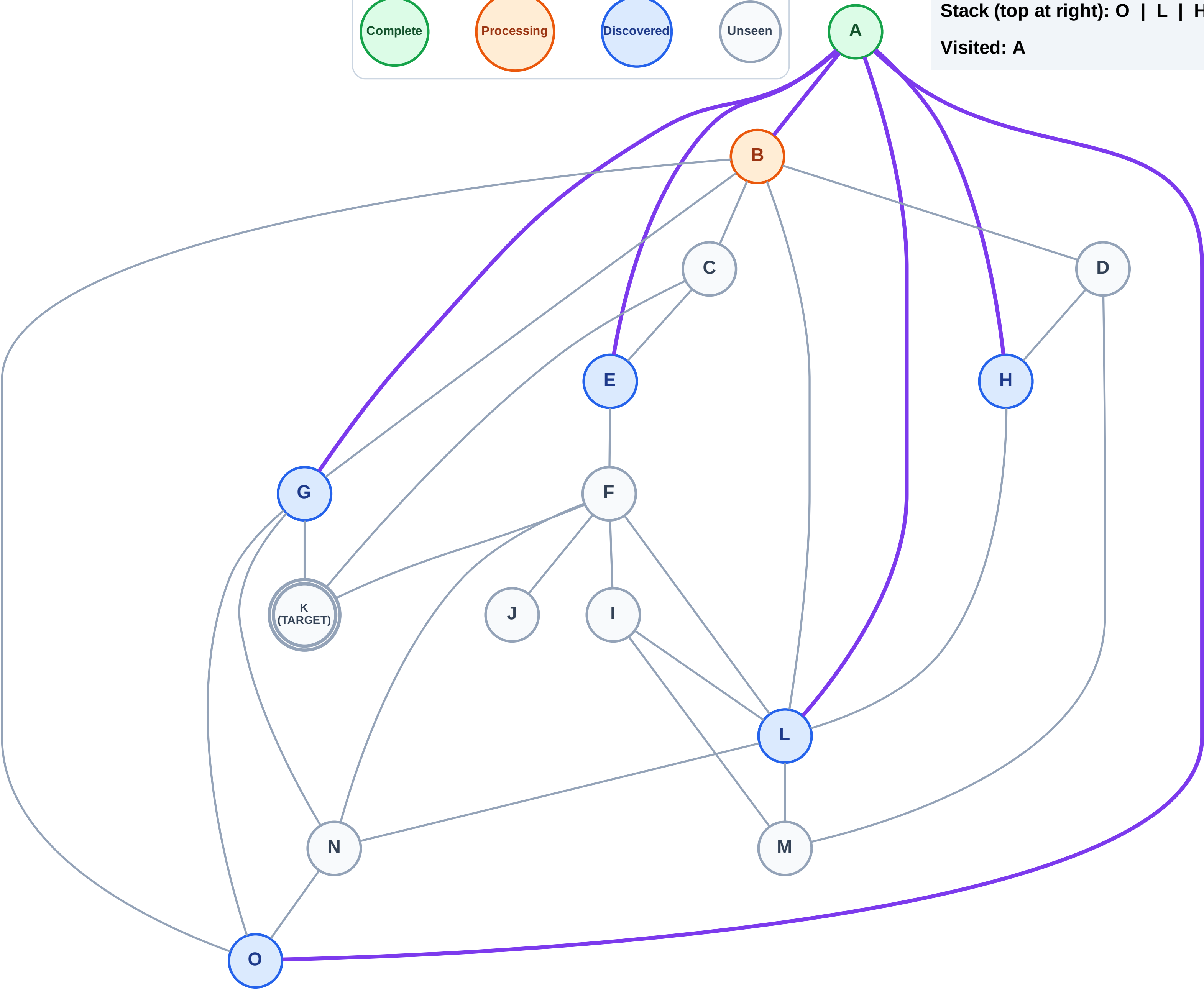
Step 4: Process node B

Legend



Stack (top at right): O | L | H | G | E

Visited: A



DFS Traversal

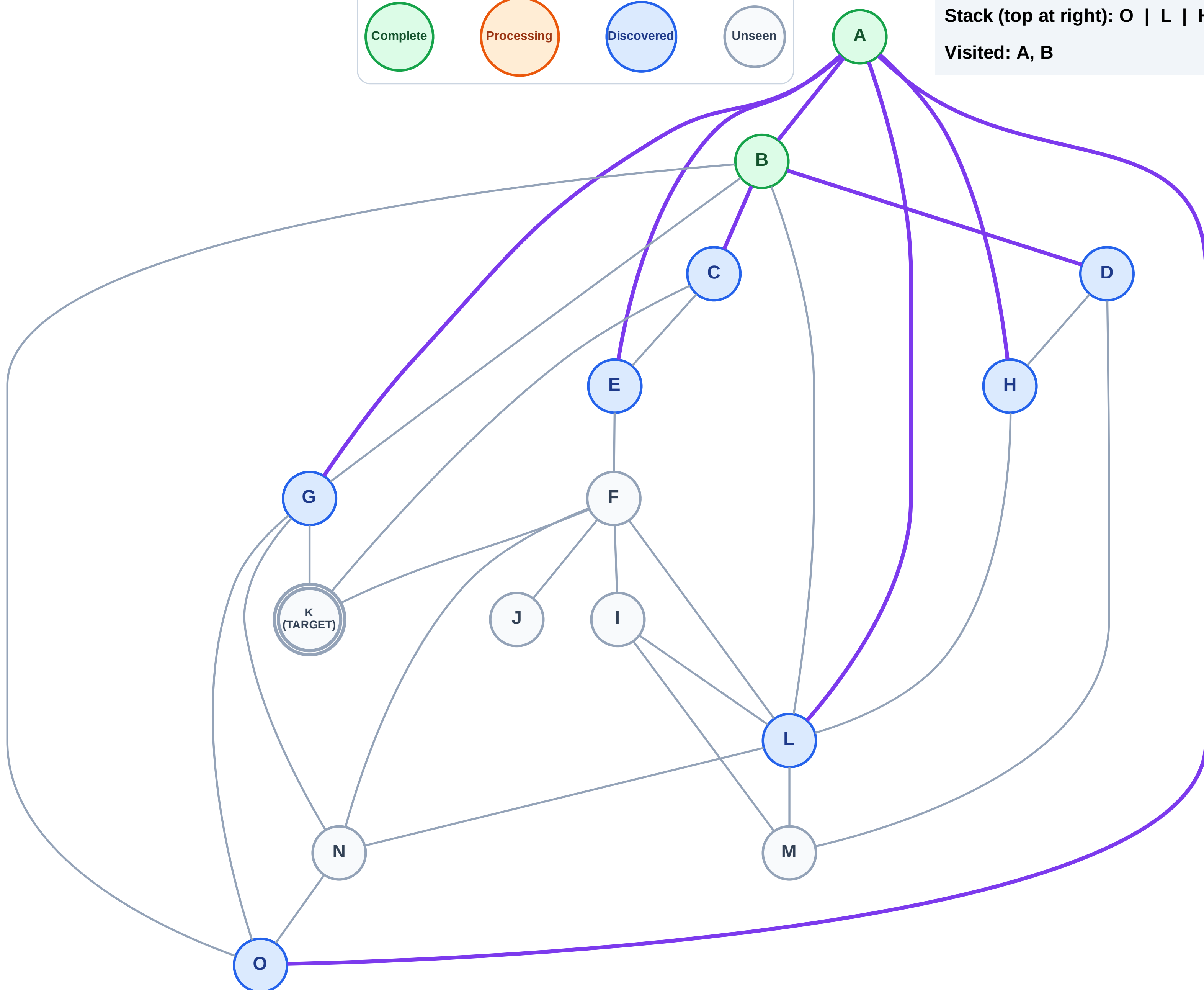
Step 5: Discover D, C from B

Legend



Stack (top at right): O | L | H | G | E | D | C

Visited: A, B



DFS Traversal

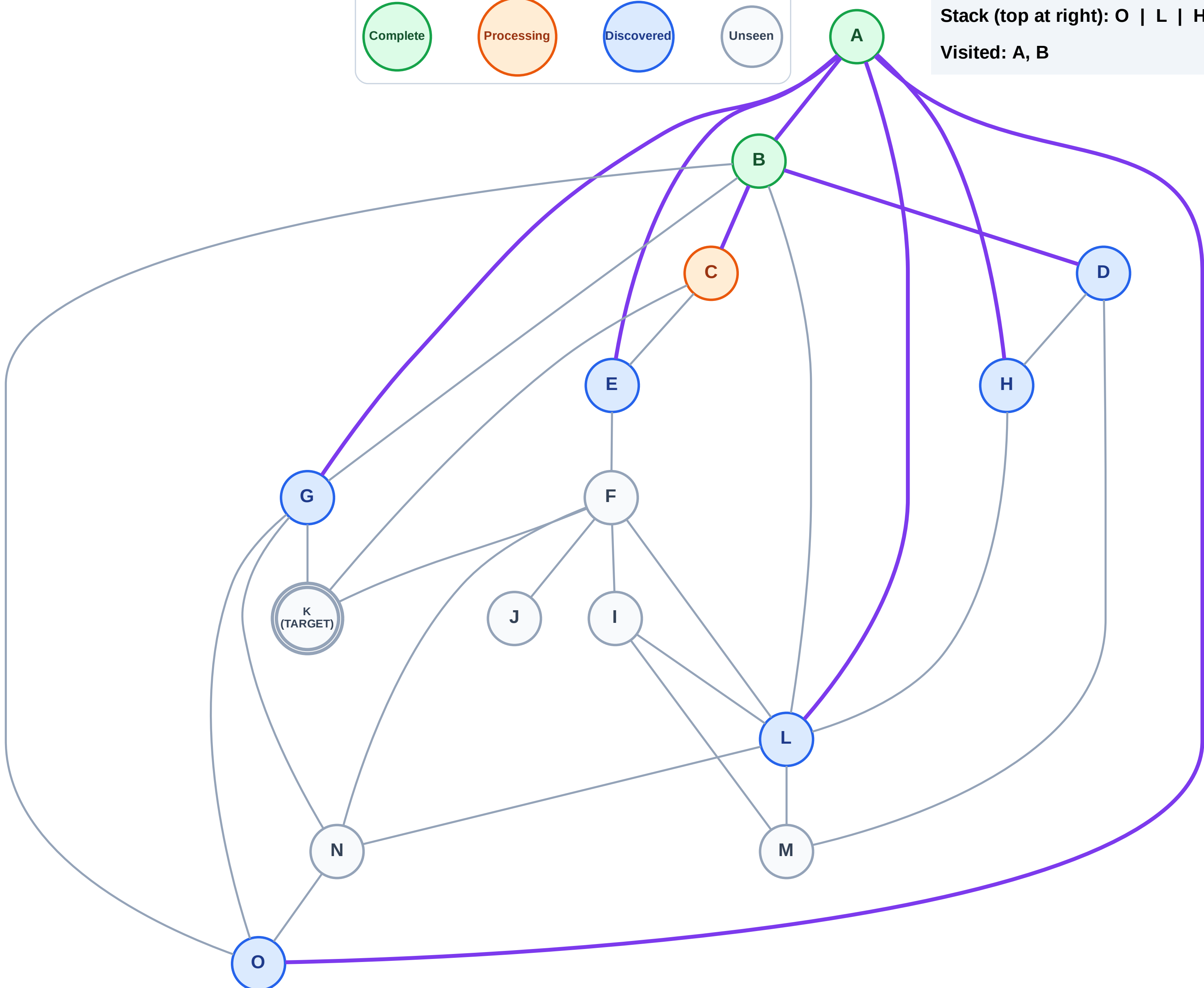
Step 6: Process node C

Legend



Stack (top at right): O | L | H | G | E | D

Visited: A, B



DFS Traversal

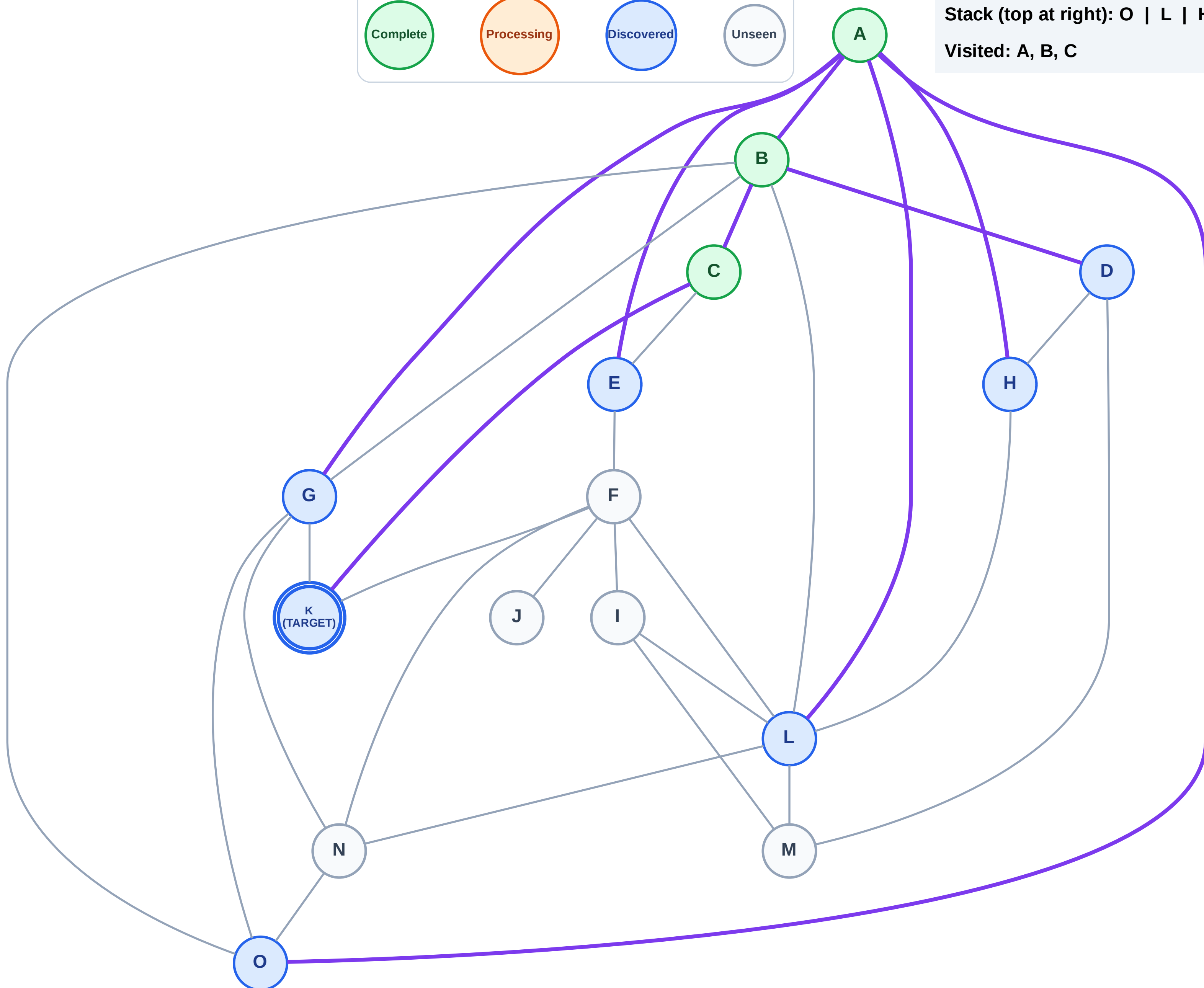
Step 7: Discover K from C

Legend



Stack (top at right): O | L | H | G | E | D | K

Visited: A, B, C



DFS Traversal

Step 8: Process node K

Legend

Complete

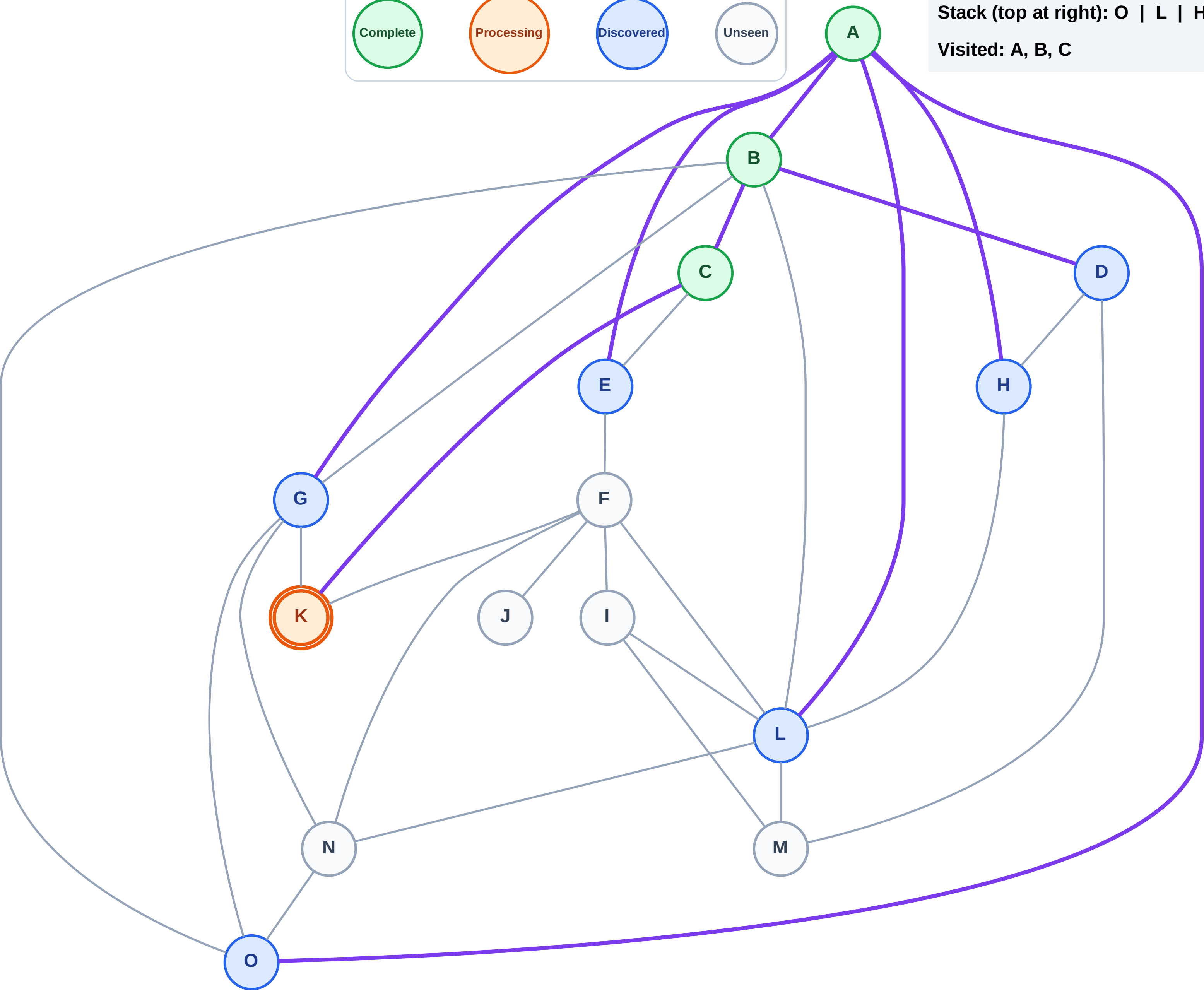
Processing

Discovered

Unseen

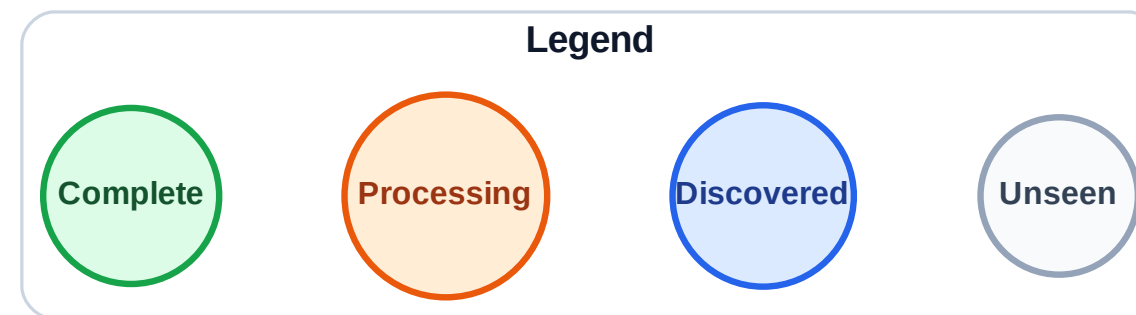
Stack (top at right): O | L | H | G | E | D

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → C → K



Stack (top at right): O | L | H | G | E | D

Visited: A, B, C, K

